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REVIEW

Micronutrient deficiency-induced oxidative stress in plants

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Abstract

Micronutrients like iron (Fe), zinc (Zn), copper (Cu), manganese (Mn), boron (B), nickel (Ni), and molybdenum (Mo) perform significant roles in the regulation of plant metabolism, growth, and development. Micronutrients, namely Fe, Zn, Cu, Mn, and Ni, are involved in oxidative stress and antioxidant defense as they are cofactors or activators of various antioxidant enzymes, viz., superoxide dismutase (Fe, Cu/Zn, Mn, and Ni), catalase (Fe), and ascorbate peroxidase (Fe). An effort has been made to incorporate recent advances along with classical work done on the micronutrient deficiency-induced oxidative stress and associated antioxidant responses of plants. Deficiency of a micronutrient produces ROS in the cellular compartments. Enzymatic and non-enzymatic antioxidant defense systems are often modulated by micronutrient deficiency to regulate redox balance and scavenge deleterious ROS for the safety of cellular constituents. ROS can strike cellular constituents such as lipids, proteins, and nucleic acids and can destruct cellular membranes and proteins. ROS might act as a signaling molecule and activate the antioxidant proteins by interacting with signaling partners such as respiratory burst oxidase homolog (RBOH), G-proteins, Ca^{2+}, mitogen activated protein kinases (MAPKs), and various transcription factors (TFs). Opinions on probable ROS signaling under micronutrient deficiency have been described in this review. However, further research is required to decipher micronutrient deficiency-induced ROS generation, perception, and associated downstream signaling events, leading to the development of antioxidant responses in plants.

Keywords Antioxidants · Lipid peroxidation · Micronutrient deficiency · Oxidative stress · ROS signaling · Reactive oxygen species

Abbreviations

AA	Ascorbate
APX	Ascorbate peroxidase
CAT	Catalase
DHA	Dehydroascorbate
DHAR	Dehydroascorbate reductase
GR	Glutathione reductase
GSH	Reduced glutathione
GSSG	Oxidized glutathione
MDHAR	Monodehydroascorbate reductase
$O_2^{\cdot-}$	Superoxide anion radical
$^{\cdot}OH$	Hydroxyl radical
ROS	Reactive oxygen species
SOD	Superoxide dismutase
TFs	Transcription factors

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Introduction

ROS are not always harmful, as they are involved in plant growth and development. A disparity between ROS generation and their associated detoxification processes causes a metabolic imbalance called oxidative stress (Mittler 2017). In other words, oxidative stress is triggered by excess ROS production. ROS may vary greatly in their chemical reactivity and properties (Mittler et al. 2022). The toxicity caused by superoxide anion ($O_2^{\cdot-}$) and hydrogen peroxide (H_2O_2) depends on their abilities to initiate a series of reactions that produce deleterious hydroxyl radicals ($OH^{\cdot}$) and lipoperoxides (Cakmak 2000). Normally, only a small amount of ROS are produced in chloroplasts, mitochondria, and

peroxisomes; however, under stress their production can increase rapidly (Desikan et al. 2001). Production and scavenging mechanisms of various ROS are depicted in Fig. 1. The antioxidant defense consists of antioxidant enzymes and antioxidant molecules (Mittler 2002). The main antioxidant enzymes are superoxide dismutase (SOD), catalase (CAT), peroxidases (PODs), and the enzymes of the Halliwell–Asada pathway, such as ascorbate peroxidase (APX), dehydroascorbate reductase (DHAR), monodehydroascorbate reductase (MDHAR), and glutathione reductase (GR) (Mittler 2002) (Fig. 1). Non-enzymatic antioxidant molecules are ascorbic acid, glutathione (GSH/GSSG), flavonoids, α-tocopherol, alkaloids, and phenolics (Mittler 2017), chlorophyll derivatives (pheophytins), amino acids (histidine, cysteine, tryptophan, methionine, and threonine) (Larson 1988), high-molecular-weight antioxidants (ferritin), endogenous phytosiderophores (Tewari et al. 2021a, b) and low-molecular-weight antioxidants (nicotianamine, and carotenoids). These antioxidant molecules generally scavenge ROS produced during biological processes (Fig. 1).

Micronutrients such as iron (Fe), zinc (Zn), copper (Cu), manganese (Mn), and nickel (Ni) constitute integral parts of various antioxidant enzymes, viz., superoxide dismutase, SOD (Fe, Cu/Zn, Mn, and Ni), CAT (Fe), and APX (Fe) (Fig. 2). A deficiency of micronutrients may result in the downregulation of the activity of single or several antioxidant enzymes. Decreased antioxidant enzymes result in the buildup of ROS in the cells (Fig. 3), in different plants under different nutritional stresses, which consequently causes oxidative damage (Tewari et al. 2021a, b). Unlike other micronutrients, boron (B) is unknown to be a constituent, activator, or inhibitor of any enzymes (Broadley et al. 2012). However, B is involved in the cross-linking of the borate ester of the pectic polysaccharide, namely rhamnogalacturonan II, a constituent of the primary cell wall, as reported by O'Neill et al. (2001). B also forms a precursor of lignin and is believed to be important for the transport of sugars (O'Neill et al. 2001). Therefore, when the concentration of micronutrient elements falls below the threshold range, either in the growth medium or in plant tissues, their deficiency symptoms might become perceptible in the plant organs. The essential micronutrients such as Fe, Zn, Cu, Mn, Mo, B, Ni and Cl have concentrations < 1000 µg/g dry matter (Kirkby 2023). However, threshold deficiency

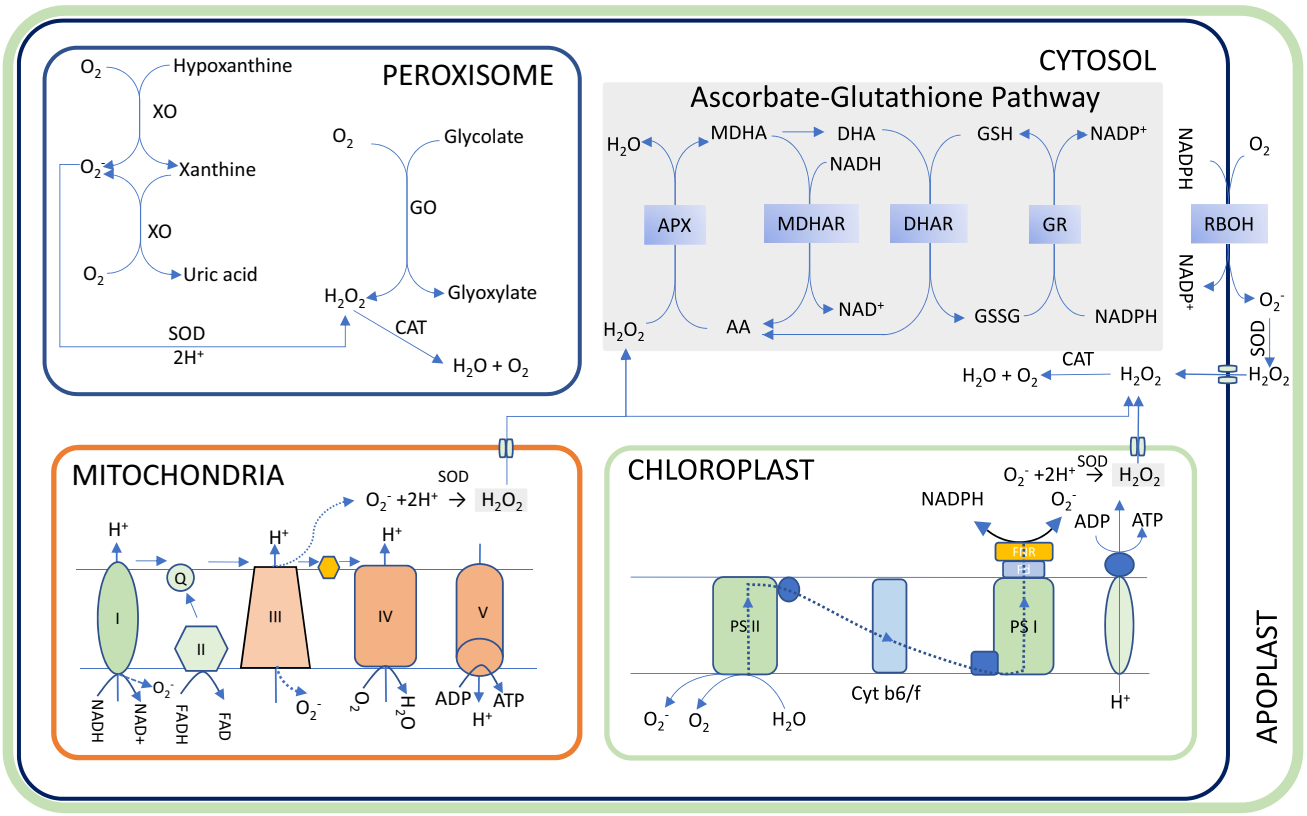


Fig. 1 Schematic diagram showing ROS production and scavenging in different subcellular compartments of plants. ROS are produced by RBOH (NADPH oxidases), leakage of electron from oversaturated ETS in chloroplasts and mitochondria. ROS are also produced by xanthine oxidase (XO) and glycolate oxidase (GO) in the peroxisome. Leakage of electron from ETS component produce O_2^- which may subsequently dismutated to H_2O_2 by SOD and H_2O_2 is scavenged by CAT, POD, and enzymes of ascorbate–glutathione pathway

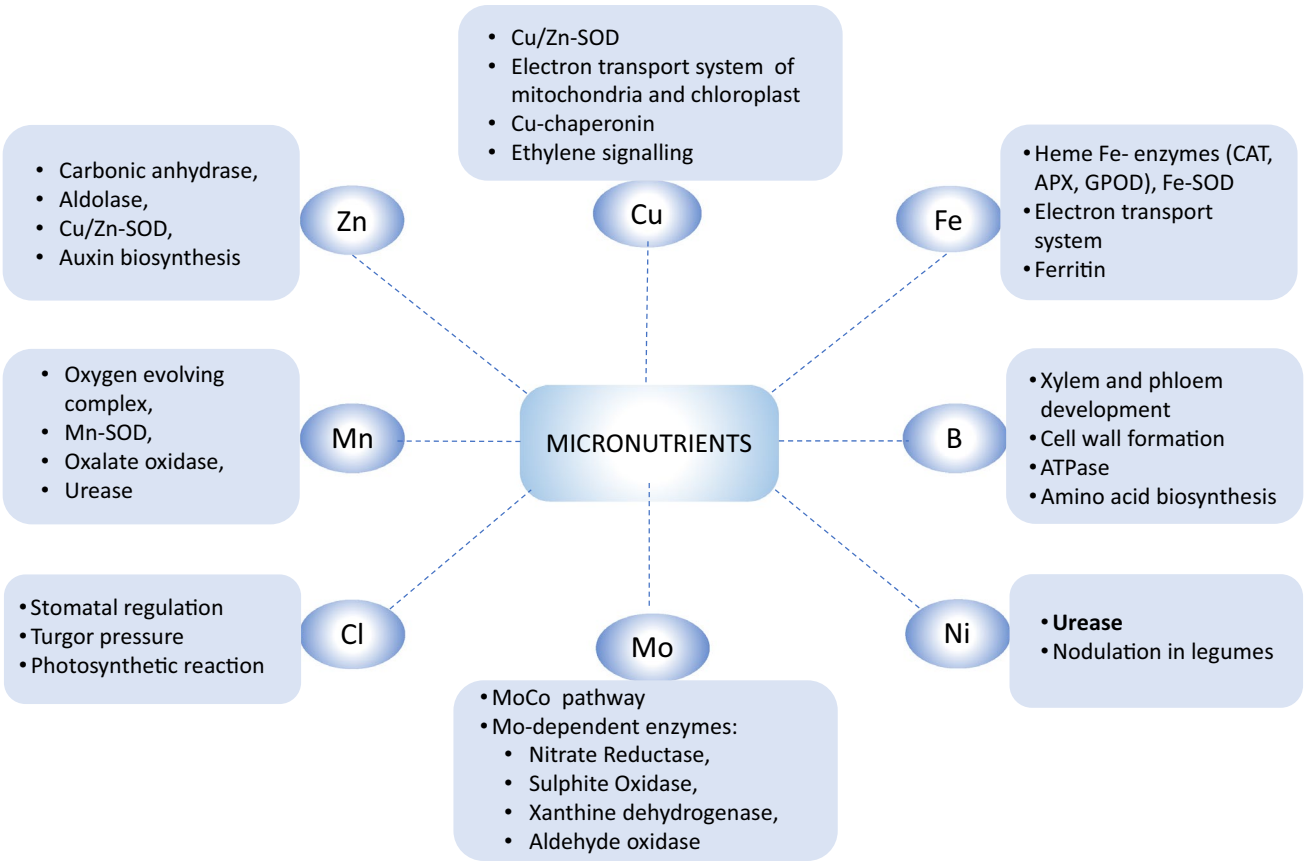


Fig. 2 Role of essential micronutrients in plants. Micronutrient has important functions in different cellular compartment and maintain cellular homoeostasis by involving themselves as an integral component with specific proteins/enzymes

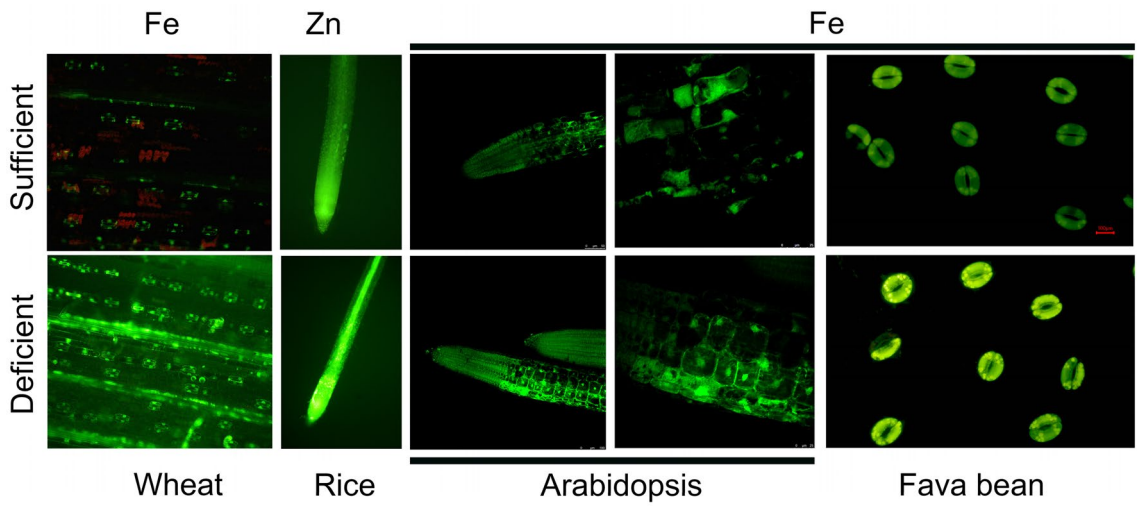


Fig. 3 Micronutrient deficiency-induced ROS production in different plant species and their organs (Wheat: epidermal peel, Rice: root apex, *Arabidopsis:* root apex, and Fava bean: lower epidermal peel)

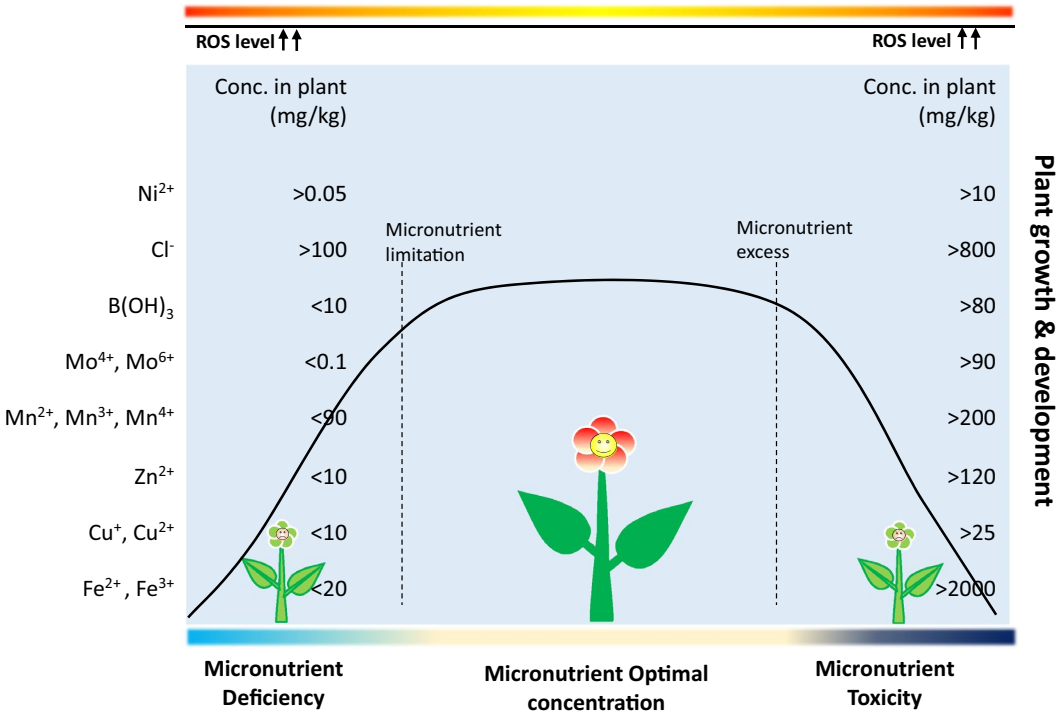


Fig. 4 Plant exhibits micronutrient deficiency and toxicity under different concentration range. These concentration range affect plant growth and development due to production of ROS in plants. The figure shows threshold concentration range (mg/kg) under deficiency and toxicity with respect to their biologically relevant ionic forms (Ni^{+2}, Cl^{-}, $B(OH)_3$, Mo^{4+}, Mo^{6+}, Mn^{2+}, Mn^{3+}, Mn^{4+}, Zn^{2+}, Cu^{+}, Cu^{2+}, Fe^{2+}, Fe^{3+}) in plants (adapted with modification from Marschner (2012) *Marschner's Mineral Nutrition of Higher Plants*)

and excess of various micronutrients have been depicted in Fig. 4. Deficiencies of mineral nutrients augment toxic ROS production in many organisms, along with affecting several physiological and metabolic pathways (Tewari et al. 2006). Several reviews are available on the oxidative stress and antioxidant responses of plants under biotic and abiotic stresses (Alscher et al. 1997). The physiological functions of the essential micronutrients linked with the deficiency symptoms in plants has been presented in a review article by Lilay et al (2024). The proteomic and genomic changes due to micronutrient deficiency and toxicity in plants were discussed by Prusty et al. (2022). Ahmed et al. (2024) showed micronutrients and their effects on crop productivity, quality and sustainability. Moreover, a book chapter enlightens the oxidative stress under some micronutrients such as Fe, Mn, Cu, and Zn deficiency in plants (Blasco et al. 2018). Another article written by Hajiboland (2012) described the effects of micronutrient deficiencies under different abiotic stresses. However, a comprehensive review of micronutrient deficiency-induced ROS production and antioxidant responses in plants is still a matter of concern. Moreover, we have tried to incorporate a timeline of important discoveries related to oxidative stress and different nutritional deficiencies, which are depicted in Table 1. In-depth review enlightening oxidative stress and antioxidant responses in plants under different micronutrient deficiencies is utmost desirable in the notion of realizing the occurrence of oxidative stress is the prime cause of plant damage and crop failure. Hence, in this review, we emphasized the causes and consequences of enhanced ROS generation and antioxidant responses in plants experiencing micronutrient deficiencies.

Micronutrient deficiency induced oxidative stress

Deficiencies in micronutrient elements modify the photosynthetic process, either by lowering the electron carriers involved in the ETS of chloroplasts (Fe, Mn, and Cu), retarding CO_2 uptake resulting in stomatal closure (Zn), or decreasing the efficiency of the C_3 cycle (Mn), which ultimately leads to increased production of reducing power, NADPH. This condition leads to the transfer of electrons to O_2 from the over-reduced carriers, initiating the O_2^- production and many other ROS, even in moderate or low light conditions (Grossman and Takahashi 2001). Limited availability of main micronutrients, which are integral constituents of antioxidant enzymes, viz., SOD (Fe, Zn/Cu, or Mn), CAT (Fe), and APX (Fe), results in enhanced ROS production in plants (Mittler 2006; Tewari et al. 2005, 2006, 2008).

Table 1 Timeline of important discoveries on oxidative stress and plant nutrition

Year	Discoveries	Refs.
1776	Oxygen and its nature	(Priestley 1776)
1876	Fenton reaction	(Fenton 1876)
1901	Oscar Loew discovered catalase in plants	(Anonymous 2001)
1922	Manganese essentiality	(McHargue 1922)
1923	Boron essentiality	(Warington 1923)
1925	Glutathione	(Hopkins 1925)
1926	Zinc essentiality	(Sommer and Lipman 1926)
1934	Haber–Weiss reaction	(Haber and Weiss 1934)
1939	Copper, Molybdenum, Nickel, Silicon, Iron, Chlorine	(Arnon and Stout 1939)
1962	Glutathione peroxidase activity	(Mills 1962)
1969	SOD activity, Identification of O_2^-	(McCord and Fridovich 1969)
1985	Introduced the term oxidative stress	(Sies 1985)
1993	Siderophore as Fe carrier	(Ishimaru and Loper 1993)
1994	Fe-SOD in plants	(Camp et al. 1994)
1995	Zn in stomatal opening in cauliflower	(Sharma et al. 1995)
1998	RBOH in plants	(Keller et al. 1998)
2000	Role of Zn in plant protection form ROS	(Cakmak 2000)
2002	Role of NO in internal Fe availability in plants	(Graziano et al. 2002)
2004	Macronutrient deficiency and antioxidant responses	(Tewari et al. 2004)
2004	Cu deficiency induced Fe-SOD in *Matteuccia struthiopteris*	(Murao et al. 2004)
2004	Zn deficiency induces oxidative stress in plants	(Sharma et al. 2004)
2004	Mn deficiency induced oxidative stress in tomato	(Shenker et al. 2004)
2005	Ca signaling	(Gilroy and Trewavas 2001)
2009	Molecular basis of Fe deficiency	(Forner-Giner et al. 2009)
2010	Ni-SOD structure and function	(Ryan et al. 2010)
2011	RBOH and ROS signaling	(Suzuki et al. 2011)
2012	Cu chaperone antioxidant protein1 in Cu homeostasis	(Shin et al. 2012)
2013	Redox control of plant growth and development	(Kocsy et al. 2013)
2013	Fe deficiency induced programmed cell death in plants	(Tewari et al. 2013a, b)
2014	Oxidative stress resistance by expression Zn deficiency transcription factor BdbZIP10	(Glover-Cutter et al. 2014)
2014	Polarization of IRT1 plays crucial role in metal homeostasis	(Barberon et al. 2014)
2015	B deficiency inhibits root cell elongation via ROS-dependent pathway	(Camacho-Cristóbal et al. 2015)
2016	Fe in stomatal closure	(Sharma et al. 2016)
2017	Molecular link between auxin and ROS-mediated polar growth	(Mangano et al. 2017)
2018	ROS in plant signaling	(Waszczak et al. 2018)
2019	B deficiency increases cytosolic Ca^{2+}	(Quiles-Pando et al. 2019)
2020	H_2O_2 sensor HPCA1	(Wu et al. 2020a, b)
2020	Autophagy increases Zn bioavailability to avoid oxidative stress	(Shinozaki et al. 2020)
2021	Evidence for a role of NO in Fe homeostasis in plants	(Tewari et al. 2021a, b)
2021	ROS coordinate the transcriptional responses to Fe availability	(von der Mark et al. 2021)
2022	Signaling by reactive molecules and antioxidants	(Minguillón et al. 2022)
2023	H_2O_2, NO, and H_2S networks in root development and signalling	(Mukherjee and Corpas 2023)
2023	Zn deficiency affects the S-nitrosoglutathione reductase and NO signaling	(Oláh et al. 2023)
2024	H_2O_2 is required for light-induced stomatal opening	(Shi et al. 2024)
2024	Superoxide signalling and antioxidant processing in the plant nucleus	(Karpinska and Foyer 2024)

Oxidative stress may become principally stern under micronutrient deficiency, when plants get constantly exposed to unfavorable climatic conditions as they require O_2 for their respiration and generate it through photosynthesis (Mittler 2006).

Iron deficiency and oxidative stress

Being an essential mineral nutrient, iron (Fe) is an intrinsic component of numerous proteins needed for diverse functions, viz., photosynthesis, respiration, and cell division (Broadley et al. 2012). It plays a key role in redox reactions in biological systems due to its chemical properties. However, it can produce ROS while existing in a free state or in large quantities (Kobayashi and Nishizawa 2012). Among all essential micronutrients, Fe is required in the highest amount in plants (Kobayashi and Nishizawa 2012). Thus, plants often exhibit Fe deficiency symptoms due to the unavailability of Fe in soil solutions. Fe availability decreases in alkaline and calcareous soils (Kobayashi and Nishizawa 2012). However, under situations that support acidification (waterlogged anaerobic soil), Fe may become available in toxic concentrations (Jeong and Guerinot 2009). The average concentration of Fe in plant tissues is approximately 100 µg g^{-1} (2 µmol g^{-1}) of dry mass. Fe (II) is active and preferable as compared to Fe (III) state, but it may differ on the plants (Strategies I and II) (Jeong and Guerinot 2009). Dicots and non-grass monocot plants utilize strategy I. These plants are characterized by Fe (III) reduction into Fe (II) by inducible membrane reductase, acidification of the rhizosphere, the release of reducing compounds (phenolics), and organic acid accumulation, namely citrate and malate, in the roots. While monocot grasses utilize Strategy II, these plants chelate Fe (III) by releasing phytosiderophores [Fe (III)-specific ligands of low molecular mass termed non-proteinogenic amino acids] (Jeong and Guerinot 2009).

Interveinal chlorosis in the young and developing leaves is a characteristic feature of Fe-deficient plants (Gupta et al. 2024; Tewari 2019). In PSI, Fe-S electron carrier enables the electron transport without NADPH production. This avoids over-energization of C3 cycle of chloroplast stroma (Zhang et al. 2023). Fe deficiency causes hindrance in electron transport due to decreased components of ETS, leading to oversaturation of ETS components. This reduced environment causes high-potential electrons to pass on to O_2 and generate O_2^- and H_2O_2 consequently (Tewari et al. 2021a, b). Many studies have already shown an enhanced production of O_2^- and H_2O_2 (Fig. 3) in various tissues/cells of Fe-deficient plants (Tewari et al. 2013a, b, 2021a; Tewari et al. 2015). Thus, Fe-deficient plants could be more susceptible to oxidative stress as Fe is an ingredient of several antioxidant enzymes involved in the detoxification of H_2O_2 (CAT, APX and POD) and the O_2^- dismutation (Fe-SOD) (Broadley et al. 2012). Fe deficiency decreases H_2O_2 decomposing enzyme activities: CAT (Iturbe-Ormaetxe et al. 1995; Tewari et al. 2005; Tewari et al. 2013a, b), APX (Iturbe-Ormaetxe et al. 1995; Zaharieva et al. 2004; Ranieri et al. 2001; Tewari et al. 2013a, b; Tewari 2019) and POD (Iturbe-Ormaetxe et al. 1995; Ranieri et al. 2001; Tewari et al. 2013a, b; Tewari 2019). Sometimes, the heme-Fe enzyme activities CAT and POD showed either an insignificant increase or no effect in the roots due to Fe deficiency (Santos et al. 2019b, a). Moreover, antioxidant enzymes of the AA-GSH cycle—MDHAR, DHAR, and GR—showed no significant effects in pea plants due to Fe deficiency (Iturbe-Ormaetxe et al. 1995). However, GR activity was found to be enhanced in Fe-deficient barley, wheat, maize (Bashir et al. 2007), and soybean (Santos et al. 2019b, a) plants. MDHAR, DHAR and GR do not require Fe as a cofactor, therefore, their activities may or may not be affected by Fe deficiency to regulate redox status of the cells.

Poor availability of catalytic Fe, which takes part in the generation of $OH^{\cdot}$ radicals, prevents Fe-deficient plants from oxidative damage (Becana and Klucas 1992; Iturbe-Ormaetxe et al. 1995; Ranieri et al. 2001). Fe-deficient plants generally exhibit a low level of lipid peroxidation compared to well-supplied Fe controls (Tewari et al. 2013a, b; Tewari et al. 2005, 2010). Low levels of tissue Fe suggest decreased Fe-SOD activity in Fe-deficient plants (Page et al. 2012; Santos et al. 2019b, a). The critical role of Fe-SOD (FSD1) in oxidative stress tolerance had been deciphered in *Arabidopsis* knockout mutants *fsd1-1* and *fsd1-2,* which were hypersensitive to oxidative stress (Melicher et al. 2024). Sevilla et al. (1984), on the other hand, reported decreased Cu/Zn-SOD in Fe-deficient citrus, while increased Cu/Zn-SOD activity has been shown by several workers under Fe-deficient conditions (Iturbe-Ormaetxe et al. 1995; Page et al. 2012; Santos et al. 2019b, a). These studies supported a compensatory rise in another SOD isoform expression when one SOD isoform expression was found to be decreased (Table 2).

A low GSSG/GSH ratio, unaffected status of GSH, malondialdehyde, α-tocopherol, and carbonyl contents, as shown by Iturbe-Ormaexte et al. (1995), do not imply the initiation of oxidative damage in Fe-deficient plants. While an enhanced DHA/AA ratio suggests oxidative stress, an enhanced total AA suggests an enhanced synthesis of AA in plants suffering from Fe deficiency (Tewari et al. 2005; Zaharieva and Abadía 2003). Being a reductant, enhanced synthesis of AA may increase functional Fe availability by reducing the unavailable pool of cellular Fe. Poor availability of functional Fe and stimulation of the antioxidant defense in Fe-deficient plants appear to be key domains that provide defense against oxidative stress (Table 2). Even though considerable evidence suggests that Fe-deficient plants have

Table 2 Effect of iron deficiency on different measures of oxidative metabolism in different plants and their parts

Parameter	Plant name	Plant part	Effect	Refs.
Reactive oxygen/nitrogen species				
Superoxide radical (O_2^-)	*Brassica napus*	Leaves	Increase	(Tewari et al. 2015; Tewari et al. 2013a, b)
	Zea mays	Leaves	Increase	(Tewari et al. 2005)
Hydrogen peroxide (H_2O_2)	*Zea mays*	Leaves	Increase	(Kumar et al. 2010; Tewari 2019; Tewari et al. 2005)
	Brassica napus	Leaves	Increase	(Tewari et al. 2015; Tewari et al. 2013a, b)
	Morus alba	Leaves	Increase	(Tewari et al. 2005)
	Brassica oleracea var. botrytis	Leaves	Increase	(Tewari et al. 2005)
	Helianthus annuus	Leaves	Increase	(Ranieri et al. 2001)
	Arabidopsis	Seedlings	Decrease	(Naranjo-Arcos et al. 2017)
	Arabidopsis	Roots	Increase	(von der Mark et al. 2021)
Nitric oxide (NO)	*Arabidopsis thaliana*	Roots	Increase	(Chen et al. 2010)
	Solanum lycopersicum	Roots	Increase	(Graziano and Lamattina 2007; Jin et al. 2009)
	Lupinus albus	Roots	Increase	(Meng et al. 2012)
Oxidative damage				
Lipid peroxidation (LPX)	*Pisum sativum*	Leaves	NS	(Iturbe-Ormaetxe et al. 1995)
	Brassica napus	Leaves	Decrease	(Tewari et al. 2015; Tewari et al. 2013a, b)
	Zea mays	Leaves	Decrease	(Kumar et al. 2010; Tewari 2019; Tewari et al. 2005)
	Morus alba	Leaves	Decrease	(Tewari et al. 2005)
	Brassica oleracea var. botrytis	Leaves	Decrease	(Tewari et al. 2005)
	Glycine max (efficient)	Leaves	NS	(Santos et al. 2019a, b)
	Glycine max (efficient)	Roots	NS	(Santos et al. 2019a, b)
Antioxidants				
AA	*Brassica napus*	Leaves	NS	(Tewari et al. 2013a, b)
	Beta vulgaris	Roots	Increase	(Zaharieva et al. 2004; Zaharieva and Abadía 2003)
	Cucumis sativus	Roots	Increase	(Zaharieva et al. 1999)
DHA	*Cucumis sativus*	Roots	Increase	(Zaharieva et al. 1999)
DHA/AA ratio	*Brassica napus*	Leaves	Increase	(Tewari et al. 2013a, b)
	Cucumis sativus	Roots	Increase	(Zaharieva et al. 1999)
GSH	*Pisum sativum*	Leaves	NS	(Iturbe-Ormaetxe et al. 1995)
	Hordeum vulgare	Leaves	NS	(Astolfi et al. 2012)
	Hordeum vulgare	Roots	Increase	(Astolfi et al. 2012)
	Beta vulgaris	Roots	Increase	(Zaharieva et al. 2004; Zaharieva and Abadía 2003)
GSSG/GSH ratio	*Pisum sativum*	Leaves	Decrease	(Iturbe-Ormaetxe et al. 1995)
	Beta vulgaris	Roots	Decrease	(Zaharieva et al. 2004; Zaharieva and Abadía 2003)
Enzyme activity				
SOD	*Pisum sativum*	Leaves	Increase	(Iturbe-Ormaetxe et al. 1995)
	Zea mays	Leaves	Increase	(Kumar et al. 2010; Tewari 2019; Tewari et al. 2005)
	Morus alba	Leaves	Increase	(Tewari et al. 2005)
	Brassica oleracea var. botrytis	Leaves	Increase	(Tewari et al. 2005)
	Brassica napus	Leaves	Increase	(Tewari et al. 2015; Tewari et al. 2013a, b)
	Glycine max (inefficient)	Leaves	Increase	(Santos et al. 2019b, a)
Cu/Zn-SOD	*Pisum sativum*	Leaves	Increase	(Iturbe-Ormaetxe et al. 1995)
Fe-SOD	*Glycine max (efficient)*	Leaves	NS	(Santos et al. 2019b, a)
	Glycine max (inefficient)	Leaves	Decrease	(Santos et al. 2019b, a)
	Glycine max (efficient)	Roots	NS	(Santos et al. 2019b, a)

Table 2 (continued)

Parameter	Plant name	Plant part	Effect	Refs.
	Glycine Max (inefficient)	Roots	Decrease	(Santos et al. 2019b, a)
Mn-SOD	*Pisum sativum*	Leaves	Increase	(Iturbe-Ormaetxe et al. 1995)
CAT	*Hordeum vulgare*	Leaves	Decrease	(Agarwala et al. 1977)
	Zea mays	Leaves	Decrease	(Kumar et al. 2010; Tewari 2019; Tewari et al. 2005)
	Poncirus trifoliata	Leaves	Decrease	(Forner-Giner et al. 2009)
	Pisum sativum	Leaves	Decrease	(Iturbe-Ormaetxe et al. 1995)
	Morus alba	Leaves	Decrease	(Tewari et al. 2005)
	Brassica oleracea var. botrytis	Leaves	Decrease	(Tewari et al. 2005)
	Glycine max (inefficient)	Leaves	NS	(Santos et al. 2019b, a)
	Glycine max (efficient)	Leaves	NS	(Santos et al. 2019b, a)
	Glycine max (inefficient)	Roots	Increase	(Santos et al. 2019b, a)
	Glycine max (efficient)	Roots	NS	(Santos et al. 2019b, a)
POD	*Hordeum vulgare*	Leaves	Decrease	(Agarwala et al. 1977)
	Zea mays	Leaves	Decrease	(Kumar et al. 2010; Tewari 2019; Tewari et al. 2005)
	Poncirus trifoliata	Leaves	Decrease	(Forner-Giner et al. 2009)
	Helianthus annuus	Leaves	Decrease	(Ranieri et al. 2001)
	Pisum sativum	Leaves	Decrease	(Iturbe-Ormaetxe et al. 1995)
	Glycine max (inefficient)	Leaves	Increase	Santos et al. (2019b, a)
	Glycine max (efficient)	Leaves	NS	(Santos et al. 2019b, a)
	Glycine max (inefficient)	Root	NS	(Santos et al. 2019b, a)
	Glycine max (inefficient)	Root	Increase	(Santos et al. 2019b, a)
APX	*Pisum sativum*	Leaves	Decrease	(Iturbe-Ormaetxe et al. 1995)
	Zea mays	Leaves	Decrease	(Kumar et al. 2010; Tewari 2019; Tewari et al. 2005)
	Morus alba	Leaves	Decrease	(Tewari et al. 2005)
	Brassica oleracea var. botrytis	Leaves	Decrease	(Tewari et al. 2005)
	Helianthus annuus	Leaves	Decrease	(Ranieri et al. 2001)
	Glycine max (inefficient)	Leaves	NS	(Santos et al. 2019b, a)
	Glycine max (efficient)	Leaves	Decrease	(Santos et al. 2019b, a)
	Glycine max (inefficient)	Roots	NS	(Santos et al. 2019b, a)
	Glycine max (efficient)	Roots	NS	(Santos et al. 2019b, a)
	Beta vulgaris	Roots	Decrease	(Zaharieva et al. 2004; Zaharieva and Abadía 2003)
	Cucumis sativus	Roots	Decrease	(Zaharieva et al. 1999)
MDHAR	*Pisum sativum*	Leaves	Decrease	(Iturbe-Ormaetxe et al. 1995)
DHAR	*Pisum sativum*	Leaves	Decrease	(Iturbe-Ormaetxe et al. 1995)
GR	*Pisum sativum*	Leaves	NS	(Iturbe-Ormaetxe et al. 1995)
	Glycine max (inefficient)	Leaves	Increase	(Santos et al. 2019b, a)
	Glycine max (efficient)	Leaves	Increase	(Santos et al. 2019b, a)
	Glycine max (inefficient)	Roots	NS	(Santos et al. 2019b, a)
	Glycine max (efficient)	Roots	NS	(Santos et al. 2019b, a)
	Beta vulgaris	Roots	Increase	(Zaharieva and Abadía 2003)
Proteome				
Cu/Zn-SOD	*Medicago sativa*	Roots	Decrease	(Rodríguez-Celma et al. 2011)
Mn-SOD	*Medicago sativa*	Roots	Increase	(Rodríguez-Celma et al. 2011)
MDHAR	*Medicago sativa*	Roots	Decrease	(Rodríguez-Celma et al. 2011)
APX	*Medicago sativa*	Roots	Decrease	(Rodríguez-Celma et al. 2011)
POD	*Medicago sativa*	Roots	Decrease	(Rodríguez-Celma et al. 2011)

Table 2 (continued)

Parameter	Plant name	Plant part	Effect	Refs.
	Cucumis sativus	Roots	Decrease	(Donnini et al. 2010)
Transcripts				
Fe-SOD	*Chlamydomonas reinhardtii*	Algal cell	Decrease	(Page et al. 2012)
Mn-SOD (MSD1, MSD2, MSD3)	*Chlamydomonas reinhardtii*	Algal cell	Increase	Page et al. 2012)
FSD1	*Arabidopsis thaliana*	Roots	Decrease	(Mai et al. 2016)
Fe-SOD	*Arabidopsis thaliana*	Leaves	Decrease	(Thimm et al. 2001)
	Arabidopsis thaliana	Roots	Decrease	(Thimm et al. 2001)
Peroxidase	*Arabidopsis thaliana*	Leaves	Increase	(Thimm et al. 2001)
	Arabidopsis thaliana	Roots	Decrease	(Mai et al. 2016; Thimm et al. 2001)
	Oryza sativa	Roots	Decrease	(Bashir et al. 2014)
Peroxidase 66	*Zea mays*	Roots	Decrease	(Li et al. 2014a, b)
Peroxidase 16, putative	*Solanum lycopersicum*	Roots	Decrease	(Zamboni et al. 2016)
Glutathione peroxidase	*Citrus*	Roots	Increase	(Licciardello et al. 2013)
Putative Glutathione S-transferase T5	*Solanum lycopersicum*	Roots	Increase	(Zamboni et al. 2016)
Glutathione S-transferase/peroxidase	*Solanum lycopersicum*	Roots	Increase	(Zamboni et al. 2016)
Ascorbate oxidase	*Solanum lycopersicum*	Roots	Decrease	(Zamboni et al. 2016)
APX	*Arabidopsis thaliana*	Leaves	Decrease	(Barberon et al. 2011)
sAPX	*Arabidopsis thaliana*	Roots	Decrease	(Mai et al. 2016)
Catalase	*Arabidopsis thaliana*	Leaves	Decrease	(Thimm et al. 2001)
	Arabidopsis thaliana	Roots	Decrease	(Thimm et al. 2001)
CAT1, CAT3	*Arabidopsis thaliana*	Roots	Decrease	(von der Mark et al. 2021)
CAT2	*Arabidopsis thaliana*	Roots	Increase	(von der Mark et al. 2021)
Lipoxygenase	*Arabidopsis thaliana*	Leaves	Increase	(Thimm et al. 2001)
	Arabidopsis thaliana	Roots	Decrease	(Thimm et al. 2001)

NS means non-significant effect, *MSD1*, *MSD2* and *MSD3* encode Mn-SODs, *FSD1* encodes Fe-SOD, *sAPX* encodes stromal ascorbate peroxidase

been protected from oxidative stress, controversy persists regarding oxidative stress in Fe-deficient plants. Tolerance of Fe-deficient plants against oxidative damage, despite the high ROS generation, may be due to the low abscisic acid level (Sharma et al. 2016) and the scant availability of catalytic Fe for $^{\cdot}OH$ generation and lipid peroxidation (Tewari et al. 2015).

Apart from ROS, another reactive radical and signaling molecule, NO, gets induced in the roots of Fe-deficient plants (Chen et al. 2010; Graziano and Lamattina 2007). Increased production of NO subsequently activates ferric chelate reductase (FRO), which reduces Fe^{3+} complexes and makes Fe^{2+} available for uptake by the Fe(II)-regulated transporter (IRT) in the roots (Chen et al. 2010; Graziano and Lamattina 2007). Exogenous NO supply prevents Fe chlorosis and increases the internal availability of Fe in plants (Graziano et al. 2002; Tewari et al. 2010). Contrarily, endogenous SA-induced deficiency of Fe responses in *Arabidopsis* (Shen et al. 2016); however, SA-induced production of NO in the *Arabidopsis* roots (Zottini et al. 2007). Increased Fe availability mediated by NO promoted growth, increased H_2O_2-scavenging activities of heme Fe-enzymes, CAT, APX, and POD, and enhanced lipid peroxidation (Tewari et al. 2010), while decreased SOD activity and low H_2O_2 concentrations were observed in Fe-deficient plants supplied with NO (Sun et al. 2007; Tewari et al. 2010). The possible mechanism of the NO-mediated increase in intracellular availability of Fe is probably modulating the redox status of the cell and upregulating the Fe uptake system (Chen et al. 2010; Tewari et al. 2010). Fe deficiency-induced changes in ROS levels and antioxidant defense systems comprising related antioxidant enzymes and molecules in plants and their organs are presented in Table 2. Moreover, changes in transcripts and proteome in Fe-deficient plants are presented in Table 2.

Zinc deficiency and oxidative stress

Zinc (Zn) is an essential micronutrient for all organisms and serves as an integral constituent of thousands of proteins (Broadley et al. 2007). Globally, its deficiency is among the most common in different agroclimatic zones (Broadley et al. 2007). Plants growing in arid and semiarid parts of the globe suffer from Zn deficiency. Zn deficiency is more prevalent in plants growing in alkaline soils (Broadley et al. 2007). The average concentration of Zn in the plant material is approximately 0.3 µmol g^{-1} of dry mass (20 µg g^{-1} of dry mass) (Broadley et al. 2012, 2007). Leucine-zipper (bZIP) TF gene family, bZIP19 and bZIP23 of *A. thaliana,* have been reported to control low Zn supply (Assunção et al. 2010). Moreover, HvZIP7 gets induced by Zn deficiency, especially in the vascular tissues of roots and leaves, and in the cell membrane (Tiong et al. 2014). It shows low-affinity Zn transport properties (Tiong et al. 2014).

The morphological and visible symptoms of Zn deficiency are stunted growth and little leaves of plants. Plants suffering from Zn deficiency also show interveinal chlorosis and ash-colored patches that finally become necrotic (Tewari et al. 2019). The chlorosis and necrosis observed in Zn-deficient plants get more severe at high intensities of light than in partial shade (Cakmak 2000). Recently, it has been demonstrated that autophagy (the major intracellular degradation mechanism) increases Zn availability to avoid light-mediated production of ROS under Zn deficiency (Shinozaki et al. 2020). Zn deficiency is reported to cause a reduction in the grana thylakoids and swelling of chloroplasts, both in the bundle sheath and mesophylls (Broadley et al. 2012).

Zinc works as a structural, functional, and regulatory cofactor in several plant proteins (Fig. 2). Some of the Zn-requiring enzymes are carbonic anhydrase (CA), alcohol dehydrogenase, L-lactate dehydrogenase, malate dehydrogenase, aldolase, Cu/Zn-SOD, and RNA polymerase (Broadley et al. 2012). CA catalyzes the hydration of CO_2 reversibly at pH 6.0 and above. Zn deficiency is known to decrease CA activity in cauliflower plants (Sharma et al. 1995). CA-catalyzed hydration of CO_2 is likely to be essential in the guard cells for malate synthesis (Sharma et al. 1995). Thus, being a constituent of CA, Zn regulates stomatal aperture by modulating CA activity (Sharma et al. 1995). Deceased opening of stomata is indicated by a decrease in stomatal aperture (Sharma et al. 1995), high diffusive resistance, and low transpiration in plants under Zn deficiency (Sharma et al. 1995; Tewari et al. 2019).

Cakmak (2000) has shown that Zn deficiency causes oxidative damage in plants due to excess ROS production (Fig. 3). Loss of membrane integrity is among the prime effects of Zn deficiency (Cakmak 2000; Yu et al. 1998). Increased H_2O_2 accumulation, higher levels of lipid peroxidation, and chlorosis and necrosis in the foliage are the key features of Zn-deficient plants (Tewari et al. 2008, 2019). These studies suggest a strong provocation of oxidative stress and subsequent damage in plants under Zn deficiency (Sharma et al. 2004; Tewari et al. 2008, 2019). Cakmak (2000) has observed an accentuation of oxidative damage with enhanced susceptibility of plants to elevated light intensities under Zn deficiency. Zn deficiency enhanced O_2^- generation by two means: (i) Zn is reported to inhibit NADPH oxidase by enhancing NADPH-dependent oxidase activity (Pinton et al. 1994) and (ii) by minimizing the ratio of NADP/NADPH owing to decreased CO_2 uptake and CO_2 fixation due to poorly open stomata and low CA activity (Cakmak 2000; Sharma et al. 1995). NADPH uses NADPH oxidase to form O_2^- in Zn-deficient plants. Enhanced activity of NADPH oxidase accompanied by high O_2^- levels was observed in Zn-deficient plants (Cakmak 2000; Cakmak and Marschner 1988; Kumar et al. 2022).

Zn-deficient plants are known to accumulate Fe in their foliage (Cakmak 2000). The plants under Fe deficiency exhibit similarity in that they pile up a high concentration of Zn (Barberon et al. 2014). Under Zn- and Fe-deficient supply, enhanced Fe and Zn concentrations, respectively, are substantiated by increases in *FRD3* (*ferric reductase defective 3*) (Charlier et al. 2015) and *IRT1* (Barberon et al. 2014) expression. FRD3 (an efflux transporter of Fe-citrate) and IRT1 are implicated in the homeostasis of metal (Barberon et al. 2014; Charlier et al. 2015). As elevated levels of Fe are known to catalyze homolytic fission of H_2O_2 in the 'Haber–Weiss' reaction, forming $OH^{\cdot}$, it is implied to be a main source of oxidative damage to proteins, lipids, and DNA of the cell in plants suffering from Zn deficiency (Cakmak 2000).

Further, the studies on the antioxidant properties of Zn propose the participation of Zn, directly or indirectly, in the cleansing of ROS by maintaining the required activities of antioxidant enzymes. Cu/Zn-SOD is the main enzyme accountable for the dismutation reaction of O_2^- into H_2O_2 and O_2 (Broadley et al. 2012). The Cu atom represents the catalytic center, and Zn is the structural component. Zn deficiency causes a decrease in Cu/Zn-SOD activity (Sharma et al. 2004; Tewari et al. 2008, 2019; Yu and Rengel 1999), and resupply of Zn quickly restores the enzyme activity in plants under Zn deficiency (Cakmak 2000; Sharma et al. 2004). It indicates that the Zn atom is an essential structural constituent, and Cu/Zn-SOD is an indicator of Zn deficiency tolerance (Yu and Rengel 1999). However, under Zn-deficient conditions, some other isoforms of SOD, namely Fe-SOD and Mn-SOD, are enhanced, leading to compensation of SOD status in plants under Zn deficiency (Tewari et al. 2008, 2019;

Table 3 Effect of zinc deficiency on different measures of oxidative metabolism in different plants and their parts

Parameter	Plant name	Plant part	Effect	Refs
Reactive oxygen species				
Nitric oxide (NO)	*Oryza sativa*	Roots	Increase	(Kumar et al. 2022)
Superoxide anion (O_2^-)	*Gossypium hirsutum*	Roots	Increase	(Cakmak 2000; Cakmak and Marschner 1988)
	Phaseolus vulgaris	Root	Increase	(Pinton et al. 1994)
	Oryza sativa	Leaves	NS	(Hajiboland and Beiramzadeh 2008)
	Oryza sativa	Leaves	Increase	(Frei et al. 2010)
	Oryza sativa	Roots	Increase	(Hajiboland and Beiramzadeh 2008)
Hydrogen peroxide (H_2O_2)	*Triticum aestivum*	Leaves	Increase	(Sharma et al. 2004)
	Coffea arabica	Leaves	Increase	(dos Santos et al. 2019a, b)
	Coffea arabica	Roots	Increase	(dos Santos et al. 2019b, a)
	Morus alba	Leaves	Increase	(Tewari et al. 2008)
	Zea mays	Leaves	Increase	(Tewari et al. 2019)
	Vigna mungo	Leaves	Increase	(Gupta et al. 2011)
	Solanum lycopersicum	Leaves	Increase	(Akther et al. 2020; Thiébaut et al. 2022)
	Oryza sativa	Leaves	Decrease	(Hajiboland and Beiramzadeh 2008)
	Oryza sativa	Leaves	Increase	(Frei et al. 2010)
	Oryza sativa	Roots	Increase	(Hajiboland and Beiramzadeh 2008; Kumar et al. 2022)
Oxidative damage				
Lipid peroxidation (LPX)	*Triticum aestivum*	Leaves	Increase	(Sharma et al. 2004)
	Coffea arabica	Leaves	Increase	(dos Santos et al. 2019b, a)
	Coffea arabica	Roots	Increase	(dos Santos et al. 2019b, a)
	Morus alba	Leaves	Increase	(Tewari et al. 2008)
	Zea mays	Leaves	NS	(Tewari et al. 2019)
	Vigna mungo	Leaves	Increase	(Gupta et al. 2011)
	Sorghum bicolor	Leaves	Increase	(Li et al. 2013)
	Oryza sativa	Leaves	Increase	(Hajiboland and Beiramzadeh 2008)
	Oryza sativa	Roots	Increase	(Hajiboland and Beiramzadeh 2008; Kumar et al. 2022)
Membrane leakage	*Gossypium hirsutum*	Roots	Increase	(Cakmak and Marschner 1988)
	Solanum lycopersicum	Leaves	Increase	(Akther et al. 2020)
Antioxidants				
AA	*Coffea arabica*	Leaves	Increase	(dos Santos et al. 2019b, a)
	Triticum aestivum	Leaves	Decrease	(Sharma et al. 2004)
	Oryza sativa	Roots	Decrease	(Frei et al. 2010)
DHA	*Triticum aestivum*	Leaves	Increase	(Sharma et al. 2004)
	Zea mays	Leaves	Increase	(Tewari et al. 2019)
DHA/AA	*Triticum aestivum*	Leaves	Increase	(Sharma et al. 2004)
	Zea mays	Leaves	Increase	(Tewari et al. 2019)
Total AA	*Vigna mungo*	Leaves	Decrease	(Gupta et al. 2011)
	Zea mays	Leaves	Increase	Tewari et al. 2019)
	Oryza sativa	Roots	Decrease	(Frei et al. 2010)
GSH	*Triticum aestivum*	Leaves	Increase	(Sharma et al. 2004)
	Oryza sativa	Leaves	Increase	(Frei et al. 2010)
Total glutathione	*Oryza sativa*	Leaves	Increase	(Frei et al. 2010)
GSSG	*Triticum aestivum*	Leaves	Increase	(Sharma et al. 2004)
	Zea mays	Leaves	Increase	(Tewari et al. 2019)
GSSG/GSH	*Triticum aestivum*	Leaves	Increase	(Sharma et al. 2004)
	Zea mays	Leaves	Increase	(Tewari et al. 2019)
Total non-protein thiol	*Vigna mungo*	Leaves	Increase	(Gupta et al. 2011)
Enzyme activity				
FCR	*Oryza sativa*	Roots	Decrease	(Kumar et al. 2022)

Table 3 (continued)

Parameter	Plant name	Plant part	Effect	Refs
NADPH oxidase	*Phaseolus vulgaris*	Root PM	Increase	(Pinton et al. 1994)
	Gossypium hirsutum	Roots	Increase	(Cakmak 2000; Cakmak and Marschner 1988)
Total SOD	*Coffea arabica*	Leaves	Decrease	(dos Santos et al. 2019b, a)
	Oryza sativa	Leaves	Decrease	(Hajiboland and Beiramzadeh 2008)
	Oryza sativa	Roots	Decrease	(Hajiboland and Beiramzadeh 2008)
	Oryza sativa	Roots	Increase	(Kumar et al. 2022)
	Oryza sativa	Leaves	Increase	(Frei et al. 2010)
	Coffea arabica	Roots	Decrease	(dos Santos et al. 2019b, a)
Cu/Zn SOD	*Triticum aestivum*	Leaves	Decrease	(Sharma et al. 2004)
	Triticum aestivum	leaves	Decrease	(Hacisalihoglu 2003)
	Oryza sativa	Leaves	Increase	(Frei et al. 2010)
	Vigna mungo	Leaves	Decrease	(Gupta et al. 2011)
	Sorghum bicolor	Leaves	Increase	(Li et al. 2013)
	Zea mays	Leaves	Decrease	(Tewari et al. 2019)
	Arabidopsis thaliana	Leaves	Decrease	(Shinozaki et al. 2020)
	Sorghum bicolor	Roots	Decrease	(Li et al. 2013)
Non-Cu/Zn SOD	*Triticum aestivum*	Leaves	Increase	(Sharma et al. 2004)
	Morus alba	Leaves	Increase	(Tewari et al. 2008)
	Zea mays	Leaves	Increase	(Tewari et al. 2019)
	Sorghum bicolor	Roots	Increase	(Li et al. 2013)
CuZn-SOD/non-CuZn SOD	*Triticum aestivum*	Leaves	Decrease	(Sharma et al. 2004)
	Morus alba	Leaves	Decrease	(Tewari et al. 2008)
	Zea mays	Leaves	Decrease	(Tewari et al. 2019)
CAT	*Phaseolus vulgaris*	Root PM	Increase	(Pinton et al. 1994)
	Oryza sativa	Leaves	NS	(Hajiboland and Beiramzadeh 2008)
	Oryza sativa	Leaves	Decrease	(Frei et al. 2010)
	Oryza sativa	Roots	NS	(Hajiboland and Beiramzadeh 2008)
	Oryza sativa	Roots	Increase	(Kumar et al. 2022)
	Oryza sativa	Leaves	Decrease	(Frei et al. 2010)
	Coffea arabica	Leaves	Increase	(dos Santos et al. 2019b, a)
	Coffea arabica	Roots	Increase	(dos Santos et al. 2019b, a)
	Triticum aestivum	Leaves	NS	(Sharma et al. 2004)
	Zea mays	Leaves	Increase	(Tewari et al. 2019)
	Hordeum vulgare	Leaves	Increase	(Batova et al. 2022)
	Vigna mungo	Leaves	Decrease	(Gupta et al. 2011)
POD	*Phaseolus vulgaris*	Root PM	Increase	(Pinton et al. 1994)
	Oryza sativa	Leaves	Increase	(Hajiboland and Beiramzadeh 2008)
	Oryza sativa	Leaves	Increase	(Frei et al. 2010)
	Oryza sativa	Roots	Increase	(Hajiboland and Beiramzadeh 2008)
	Triticum aestivum	Leaves	Increase	(Sharma et al. 2004)
	Vigna mungo	Leaves	Increase	(Gupta et al. 2011)
	Zea mays	Leaves	Increase	(Tewari et al. 2019)
	Morus alba	leaves	Increase	(Tewari et al. 2008)
APX	*Phaseolus vulgaris*	Root PM	Increase	(Pinton et al. 1994)
	Oryza sativa	Leaves	NS	(Hajiboland and Beiramzadeh 2008)
	Oryza sativa	Roots	Increase	(Hajiboland and Beiramzadeh 2008; Kumar et al. 2022)
	Oryza sativa	Leaves	Increase	(Frei et al. 2010)
	Coffea arabica	Leaves	Increase	(dos Santos et al. 2019b, a)
	Coffea arabica	Roots	Increase	(dos Santos et al. 2019b, a)
	Triticum aestivum	Leaves	Increase	(Sharma et al. 2004)

Table 3 (continued)

Parameter	Plant name	Plant part	Effect	Refs
	Vigna mungo	Leaves	Decrease	(Gupta et al. 2011)
	Morus alba	Leaves	Increase	(Tewari et al. 2008)
GPX	*Vigna mungo*	Leaves	Increase	(Gupta et al. 2011)
MDHAR	*Oryza sativa*	Leaves	NS	(Frei et al. 2010)
DHAR	*Oryza sativa*	Leaves	Increase	(Frei et al. 2010)
GR	*Phaseolus vulgaris*	Root PM	Increase	(Pinton et al. 1994)
	Triticum aestivum	Leaves	Increase	(Sharma et al. 2004)
	Zea mays	Leaves	Increase	(Tewari et al. 2019)
	Oryza sativa	Leaves	NS	(Hajiboland and Beiramzadeh 2008)
	Oryza sativa	Roots	NS	(Hajiboland and Beiramzadeh 2008)
	Oryza sativa	Roots	Increase	(Kumar et al. 2022)
	Oryza sativa	Leaves	Increase	(Frei et al. 2010)
Proteome				
Fe-SOD	*Solanum tuberosum*	Leaves	Increase	(Cheng et al. 2019)
CAT	*Solanum tuberosum*	Leaves	Decrease	(Cheng et al. 2019)
PR protein	*Solanum tuberosum*	Leaves	Increase	(Cheng et al. 2019)
Thioredoxin	*Zea mays*	Leaves	Increase	(Zhang et al. 2019)
Transcripts				
CSD1	*Sorghum bicolor*	Roots	Decrease	(Li et al. 2013)
CSD2	*Sorghum bicolor*	Leaves	Increase	(Li et al. 2013)
SOD1	*Triticum aestivum*	Leaves	Decrease	(Hacisalihoglu 2003)
CSD2	*Brachypodium distachyon*	Leaves	Increase	(Glover-Cutter et al. 2014)
SOD	*Zea mays*	Roots	Decrease	(Mager et al. 2018)
CAT2	*Brassica rapa*	Leaves	Increase	(Li et al. 2014a, b)
	Hordeum vulgare	Leaves	Increase	(Batova et al. 2022)
POD	*Oryza sativa*	Leaves	Increase	(Nanda et al. 2017)
	Zea mays	Roots	Decrease	(Mager et al. 2018)
GPX6	*Brachypodium distachyon*	Leaves	Increase	(Glover-Cutter et al. 2014)
SOD1	*Brachypodium distachyon*	Leaves	NS	(Glover-Cutter et al. 2014)
SOD1	*Oryza sativa*	Leaves	Decrease	(Nanda et al. 2017)
GST24	*Brachypodium distachyon*	Leaves	NS	(Glover-Cutter et al. 2014)

NS means non-significant effect, *CSD1*, *CSD2* encode Cu/Zn-SODs, *sAPX* encodes stromal ascorbate peroxidase, *GST24* encodes glutathione S-transferase

Yu and Rengel 1999). Zn deficiency is also described by increased activities of non-Zn antioxidant enzymes, viz., CAT (Kumar et al. 2022; Tewari et al. 2019), POD (Sharma et al. 2004; Tewari et al. 2019), APX (Kumar et al. 2022; Tewari et al. 2019), and GR (Kumar et al. 2022; Sharma et al. 2004). In view of increases in the activities of antioxidant enzymes, deficiency of Zn also increases the ratio of DHA/AA and GSSG/GSH, which suggests strong oxidative stress in plants (Sharma et al. 2004; Tewari et al. 2019). Zn deficiency elevates oxidative stress by increasing ROS generation, disturbed or unbalanced redox couples (DHA/AA and GSSG/GSH ratio), and increased lipid peroxidation despite upregulation in antioxidant enzyme activities (SOD, CAT, APX POD, and GR). Zn deficiency-induced changes in ROS levels and defense responses of antioxidants in various plant parts are presented in Table 3. Moreover, changes in transcripts and proteome in Zn-deficient plants are presented in Table 3.

Copper deficiency and oxidative stress

Copper (Cu) is a crucial redox-active element in living systems (Broadley et al. 2012). The average Cu composition in leaves is 0.16 µmol g^{-1} dry mass (10 µg g^{-1} dry mass). Due to its redox properties, the concentration of Cu in cells desires to be maintained at an optimal level (Yruela 2009). Cu is crucial for normal homeostasis in plants, and plants develop deficiency symptoms due to its inadequate supply, affecting young leaves and reproductive organs. Cu

deficiency symptoms are initiated in the young leaves of the plants. The plants suffering from Cu deficiency showed chlorosis, necrosis, and scorches in the leaves, starting from the margins to the apex of the young and emerging foliage, which eventually faded and dropped from the branch. Purplish-brown necrotic spots in the interveinal regions of the mature leaves have also been observed in Cu-deficient mulberry plants (Tewari et al. 2006).

Cu is a necessary constituent for many electron transfer carriers and is implicated in both the reduction and oxidation pathways of ETS in chloroplasts and mitochondria, respectively (Tewari et al. 2006). Cu^{2+} gets reduced rapidly to Cu^{+} and is unstable. Cu^{2+} often interacts with N in the side chains of histidine, while Cu^{+} interacts with the S present in cysteine or methionine (Yruela 2009). A Cu-related terminal oxidase may react with O_2 directly, where the reaction gets catalyzed by Cu (Broadley et al. 2012). It also offers safety against pathogenic infections, playing a role in the lignification of the cell wall and the metabolism of AA (Broadley et al. 2012). Utmost Cu is present in plants in complex forms (98%), and free Cu^{2+} and Cu^{+} concentrations are found to be extremely low in the cytoplasm (Broadley et al. 2012). A mild rise in Cu concentration may cause toxicity, as redox exchange and cycling between Cu^{2+} and Cu^{+} might initiate the production of the hazardous $^{\cdot}OH$ radical, leading to the damage of various biomolecules, viz., lipids, membranes, nucleic acids, and proteins (Yruela 2009).

The plants constitute a relatively lower number of Cu-dependent proteins compared to other metalloproteins (Yruela 2005, 2009). The most generous Cu-containing proteins are plastocyanin found in the chloroplasts and Cu/Zn-SOD. Cu is exploited as an important cofactor in enzymes, namely Cu/Zn-SOD, ascorbate oxidase, cytochrome c oxidase, amino oxidase, laccase, polyphenol oxidase, and plastocyanin (Yruela 2009). Being a fundamental component of Cu/Zn-SOD, Cu plays a decisive role in the removal of O_2^{-} (Tewari et al. 2006). Cu/Zn-SOD exists in three isoforms in *Arabidopsis*. Major Cu/Zn-SOD isoforms are located in the cytosolic region (CSD1), chloroplast stroma (CSD2), and peroxisome (CSD3) (Yruela 2009). These Cu/Zn-SOD isoforms dismutate potentially damaging O_2^{-} to a less detrimental H_2O_2 (Tewari et al. 2006). Superoxide anion (O_2^{-}) and H_2O_2 may produce hazardous $OH^{\cdot}$ in the Haber–Weiss reaction in the presence of Cu^{2+} (Apel and Hirt 2004). The Cu reactivity makes it a good candidate for redox activities, but it also makes Cu toxic, damaging the chloroplasts, membrane-associated lipids, and proteins (Maksymiec 1998; Pätsikkä et al. 2002). Free Cu can easily oxidize the sulfhydryl linkage of proteins, leading to functional and organizational disorders. The damage of membrane-associated lipids and ion leakage from cell membranes was observed due to increased production of ROS (Tewari et al. 2006).

The deficiency of Cu is mostly ranging from 1 to 5 μg g^{-1} dry mass, which depends on plant species (Broadley et al. 2012). Cu deficiency is visibly characterized in the young and developing leaves, while the meristems show poor Cu transport (Tewari et al. 2006). Cu deficiency reduces plant growth, alters root and leaf formation, and affects chlorophyll and photosynthesis adversely (Yruela 2005, 2009). Photosystem II, photosystem I, and the ETS have been described as being inhibited by Cu deficiency (Droppa et al. 1987). Suppressed ETS operation may cause an overflow of electrons from the ETS of mitochondria or chloroplasts to O_2 and produce O_2^{-}. The lipid destruction and ion leakage through cell membrane results from the excessive production of O_2^{-} in the wheat roots suffering from Cu deficiency (Quartacci et al. 2001) and in the leaves of mulberry plants (Tewari et al. 2006). Cu deficiency, apart from enhancing O_2^{-} generation, also improved SOD activity and the buildup of H_2O_2, suggesting the provocation of oxidative stress. Increased lipid peroxidation in the foliage of Cu-deficient plants indicates oxidative damage due to increased ROS production. Foliage of Cu-deficient mulberry plants accumulates H_2O_2 particularly in their trichomes, which could be an outcome of delayed scavenging of H_2O_2 in the trichomes (Tewari et al. 2006).

Total activity of SOD gets improved in Cu-deficient mulberry plants despite suppression in Cu/Zn-SOD (Tewari et al. 2006). This suppression of Cu/Zn-SOD in Cu-deficient plants tends to be counterbalanced by overexpressing Fe-SOD, as earlier reported by Murao et al. (2004). In other Cu-deficient species, viz., lupins, tobacco, tomato, *Arabidopsis*, mustard, and rice, a similar compensatory mechanism of enhancing Fe-SOD has been observed to cope with enhanced O_2^{-} generation (Cohu and Pilon 2007; Yu et al. 1998; Yu and Rengel 1999). Functionally equivalent Fe proteins replace Cu proteins under low concentrations of Cu, as has been reported (Yruela 2009). Fe-SOD replaces chloroplast-associated CSD2 upon Cu limitation in plants (Abdel-Ghany et al. 2005). This was linked with a decline in the corresponding Cu chaperone expression for SOD (CCS) which brings Cu to Cu/Zn-SOD (Cohu et al. 2009). CCS is active in plastids and cytosols for delivering Cu to CSD2, and for delivering Cu to both CSD1 and CSD3, respectively (Burkhead et al. 2009).

Apart from SOD activity, Cu deficiency also induced APX, GR, CAT, and POD activities. Increases in APX and GR activities in Cu-deficient plants suggest the stimulation of the AA-GSH pathway in such plants. A decline in AA and an increase in the DHA/AA ratio have been observed in Cu-deficient plants. These observations suggest the activated operation of the AA-GSH pathway to address Cu deficiency-induced oxidative stress (Tewari et al. 2006). Cu

Table 4 Effect of copper deficiency on different measures of oxidative metabolism in different plants and their parts

Parameter	Plant name	Plant part	Effect	Refs.
Reactive oxygen/nitrogen species				
Superoxide anion (O_2^-)	*Morus alba*	Leaves	Increase	(Tewari et al. 2006)
	Triticum durum	Roots	Increase	(Quartacci et al. 2001)
Hydrogen peroxide (H_2O_2)	*Triticum aestivum*	Leaves	Increase	(Agarwal et al. 2006)
	Morus alba	Leaves	Increase	(Tewari et al. 2006)
	Coffea arabica	Leaves	Increase	(Santos et al. 2017)
	Morus alba	Leaves	Increase	(Tewari et al. 2006)
	Citrus (rootstocks)	Leaves	Increase	(Hippler et al. 2016)
	Citrus (Swingle citrimelo)	Leaves	NS	(Hippler et al. 2018)
	Citrus (Swingle citrimelo)	Roots	NS	Hippler et al. 2018)
Oxidative damage				
Lipid peroxidation (LPX)	*Triticum aestivum*	Leaves	Increase	(Agarwal et al. 2006)
	Morus alba	Leaves	Increase	(Tewari et al. 2006)
	Amomum villosum	Leaves	Increase	(Feng et al. 2012)
	Coffea arabica	Leaves	Increase	(Santos et al. 2017)
	Citrus (rootstocks)	Leaves	Increase	(Hippler et al. 2016)
	Citrus (Swingle citrimelo)	Leaves	NS	(Hippler et al. 2018)
	Citrus (Swingle citrimelo)	Roots	NS	(Hippler et al. 2018)
Enzyme activity				
SOD	*Triticum aestivum*	Leaves	Decrease	(Agarwal et al. 2006)
	Amomum villosum	Leaves	Decrease	(Feng et al. 2012)
	Citrus (rootstocks)	Leaves	Increase	(Hippler et al. 2016)
	Coffea arabica	Leaves	Increase	(Santos et al. 2017)
	Coffea arabica	Roots	Increase	(Santos et al. 2017)
	Morus alba	Leaves	Increase	(Tewari et al. 2006)
	Nicotiana tabacum	Leaves	Increase	(Yu et al. 1998)
	Lupinus angustifolius	Leaves	Decrease	(Yu and Rengel 1999)
Fe-SOD	*Triticum aestivum*	Leaves	Decrease	(Agarwal et al. 2006)
	Citrus (rootstocks)	Leaves	Increase	(Hippler et al. 2016)
	Arabidopsis thaliana	Leaves	Increase	(Cohu et al. 2009; Cohu and Pilon 2007; Yamasaki et al. 2008)
	Brassica juncea	Leaves	Increase	(Cohu and Pilon 2007)
	Oryza sativa	Leaves	Increase	(Cohu and Pilon 2007)
	Nicotiana tabacum	Leaves	Increase	(Yu et al. 1998)
	Lupinus angustifolius	Leaves	Decrease	(Yu and Rengel 1999)
Mn-SOD	*Triticum aestivum*	Leaves	Decrease	(Agarwal et al. 2006)
	Citrus (rootstocks)	Leaves	Increase	(Hippler et al. 2016)
	Oryza sativa	Leaves	Increase	(Cohu and Pilon 2007)
	Zea mays	Leaves	Increase	(Cohu et al. 2009)
	Pisum sativum	Leaves	Increase	(Ayala and Sandmann 1988)
	Nicotiana tabacum	Leaves	Increase	(Yu et al. 1998)
	Lupinus angustifolius	Leaves	Decrease	(Yu and Rengel 1999)
Cu/Zn-SOD	*Triticum aestivum*	Leaves	Decrease	(Agarwal et al. 2006)
	Citrus (rootstocks)	Leaves	Decrease	Hippler et al. 2016
	Arabidopsis thaliana	Leaves	Decrease	(Cohu et al. 2009; Cohu and Pilon 2007; Yamasaki et al. 2008)
	Zea mays	Leaves	Decrease	(Cohu and Pilon 2007)
	Brassica juncea	Leaves	Decrease	(Cohu and Pilon 2007)
	Oryza sativa	Leaves	Decrease	(Cohu and Pilon 2007)
	Lycopersicon lycopersicum	Leaves	Decrease	(Cohu and Pilon 2007)

Table 4 (continued)

Parameter	Plant name	Plant part	Effect	Refs.
	Lemna gibba	Fronds	decrease	(Vaughan et al. 1982)
	Pisum sativum	Leaves	Decrease	(Ayala and Sandmann 1988; Broadley et al. 2012)
CAT	*Triticum aestivum*	Leaves	Decrease	(Agarwal et al. 2006)
	Citrus (rootstocks)	Leaves	NS	(Hippler et al. 2016)
	Coffea arabica	Leaves	Increase	(Santos et al. 2017)
	Coffea arabica	Roots	Increase	(Santos et al. 2017)
	Morus alba	Leaves	Increase	(Tewari et al. 2006)
	Citrus (Swingle citrimelo)	Leaves	Decrease	(Hippler et al. 2018)
	Citrus (Swingle citrimelo)	Roots	Decrease	(Hippler et al. 2018)
Ascorbate oxidase (AO)	*Pisum sativum*	Leaves	Decrease	(Ayala and Sandmann 1988)
APX	*Triticum aestivum*	Leaves	Decrease	(Agarwal et al. 2006)
	Citrus (rootstocks)	Leaves	NS	(Hippler et al. 2016)
	Coffea arabica	Leaves	Increase	(Santos et al. 2017)
	Coffea arabica	Roots	Increase	(Santos et al. 2017)
	Amomum villosum	Leaves	Increase	(Feng et al. 2012)
	Morus alba	Leaves	Increase	(Tewari et al. 2006)
	Citrus (Swingle citrimelo)	Leaves	NS	(Hippler et al. 2018)
	Citrus (Swingle citrimelo)	Roots	NS	(Hippler et al. 2018)
POD	*Citrus* (rootstocks)	Leaves	NS	(Hippler et al. 2016)
	Chrysanthemum morifolium	-	Decrease	(Broadley et al. 2012)
	Morus alba	Leaves	Increase	(Tewari et al. 2006)
	Citrus (Swingle citrimelo)	Leaves	NS	(Hippler et al. 2018)
	Citrus (Swingle citrimelo)	Roots	NS	(Hippler et al. 2018)
GR	*Morus alba*	Leaves	Increase	(Tewari et al. 2006)
Ascorbate				
AA	*Morus alba*	Leaves	Decrease	(Tewari et al. 2006)
DHA	*Morus alba*	Leaves	Increase	(Tewari et al. 2006)
DHA/AA	*Morus alba*	Leaves	Increase	(Tewari et al. 2006)
Proteome				
Glutathione reductase	*Brassica napus*	Chloroplast	Decrease	(Cobine et al. 2014)
Peroxiredoxin-2E	*Brassica napus*	Chloroplast	Decrease	(Cobine et al. 2014)
Thioredoxin-like protein CSDSP32	*Brassica napus*	Chloroplast	Increase	(Cobine et al. 2014)
ATX1	*Arabidopsis thaliana*	Seedling	Increase	(Shin et al. 2012)
CCH	*Arabidopsis thaliana*	Seedling	Increase	(Shin et al. 2012)
Transcripts				
Fe-SOD	*Matteuccia struthiopteris*	Prothalli	Increase	(Murao et al. 2004)
FSD1	*Arabidopsis thaliana*	Leaves	Increase	(Abdel-Ghany et al. 2005; Bernal et al. 2012; Perea-García et al. 2016; Ramamurthy et al. 2018)
CSD1	*Arabidopsis thaliana*	Leaves	Decrease	(Abdel-Ghany et al. 2005; Bernal et al. 2012; Ramamurthy et al. 2018)
CSD2	*Arabidopsis thaliana*	Leaves	Decrease	(Abdel-Ghany et al. 2005; Bernal et al. 2012; Ramamurthy et al. 2018)
FSD1	*Arabidopsis thaliana*	Roots	Increase	(Bernal et al. 2012)
FSD1	*Arabidopsis thaliana*	Shoots	Increase	(Bernal et al. 2012)
CSD1	*Arabidopsis thaliana*	Roots	Decrease	(Bernal et al. 2012)
CSD2	*Arabidopsis thaliana*	Roots	Decrease	(Bernal et al. 2012)
ROXY15 (Glutaredoxin family protein)	*Arabidopsis thaliana*	Roots	Decrease	(Bernal et al. 2012)

NS means non-significant effect *CSD1* and *CSD2* encode Cu/Zn SODs, *FSD1* encodes Fe- SOD

deficiency-induced changes in H_2O_2 and O_2^- and antioxidant defense responses of plants are presented in the following Table 4. Moreover, changes in transcripts and proteome in Cu- deficient plants are also presented in Table 4.

Manganese deficiency and oxidative stress

Manganese (Mn) is an essential micronutrient in all plants. Mn (II) is the very common form, although it oxidizes into Mn (III) and Mn (IV) forms. Mn plays a central role in redox processes (Broadley et al. 2012). Uptake of Mn^{2+} occurs through facilitated diffusion by plants. Plasma membrane-located transporters of Mn related to several gene families (ZIP family: IRT1; NRAMP family: NRAMP1, NRAMP3, NRAMP4), were associated with Mn transport in plants (Lanquar et al. 2010; Pedas et al. 2008; Tewari et al. 2013a, b). These all transporters have further been implicated in certain heavy metal transport (Fox and Guerinot 1998). Moreover, the *Arabidopsis* gene *CAX* (calcium exchangers) is reported to be involved in the transport of metal from the cytosol to the vacuole (Alejandro et al. 2020).

Plants require approximately 20–40 µg Mn g^{-1} of dry mass for their biological functions; however, many plants accumulate more Mn (30 to 500 µg Mn g^{-1} dry mass) in their tissues (Shao et al. 2017). Excess Mn is present in acidic soils with a low pH (< 5.5) and/or in reducing environmental conditions (Millaleo et al. 2013). Mn is a rather low-movable element in plants. Mn availability due to interaction with Ca^{2+} in the soil solution get decreases with increasing soil pH. Sandy soils with high porosity favor the oxidation of Mn to unavailable Mn oxide (Broadley et al. 2012; Schmidt et al. 2016). Mn deficiency symptoms initiate in the young or relatively older middle leaves of plants growing on soil deficient in Mn. Symptoms of Mn deficiency in plants are frequently observed as interveinal chlorotic leaves (Tewari et al. 2013a, b). Chloroplasts are the most sensitive cell organelles of Mn-deficient plants, and it causes disintegration in the lamellar membrane system of the chloroplasts (Alejandro et al. 2020).

The water splitting system during the light reaction of photosynthesis is the biological role of Mn in green organisms (Broadley et al. 2012; Schmidt et al. 2016). Mn also plays a role in chloroplast biogenesis (Alejandro et al. 2020; Zanão Júnior et al. 2010). Though many enzymes get activated by Mn, only some are Mn-containing enzymes, viz., the oxygen-evolving complex of photosystem (PS) II (Allen et al. 2007; Millaleo et al. 2013; Schmidt et al. 2016), the Mn-SOD, oxalate oxidase, and the purple acid phosphatase (Broadley et al. 2012). The catalytic Mn_4Ca cluster is specifically required for water-oxidizing processes in PS II (Schmidt et al. 2016). A remarkable reduction in PS II quantum yield (ΦPS II) in Mn-deficient leaves describes the loss of photochemical efficiency (Husted et al. 2009). Thus, Mn deficiency causes detrimental effects on the photosynthetic apparatus, leading to reduced photosynthetic ETS and oxidative stress (Schmidt et al. 2016). In the apoplast, oxalate oxidase plays a role in defense against pathogens by generating microcidal concentrations of H_2O_2 and causing H_2O_2-mediated lignification of the cell wall (Alejandro et al. 2020). Being a constituent of Mn-SOD, Mn^{2+} participates in the O_2^- dismutation to H_2O_2. Mn-SOD is located in the mitochondria and cytosol of the plant cell. Two other SOD forms, such as Cu/Zn-SOD and Fe-SOD, participate in the dismutation of O_2^- (Broadley et al. 2012).

Suppressed cyanide-resistant forms of SOD activities, which are comprised of Mn-SOD and Fe-SOD, have been observed in Mn-deficient tobacco plants (Yu et al. 1998, 1999). Mn-deficient *Chlamydomonas reinhardtii* also exhibits the loss of Mn-SOD activity despite multifold upregulation of the genes *MSD3–MSD5*, encoding various forms of the Mn-SODs, in Mn-deficient cells (Allen et al. 2007). This might suggest the critical necessity of Mn^{2+} for Mn-SOD. However, Mn-deficient mulberry does not suppress Mn-SOD but enhances Fe-SOD (Tewari et al. 2013a, b), resulting either from prioritized incorporation of the Mn in its respective apoproteins or from over-expression of Fe-SOD (Tewari et al. 2013a, b). Generally, Fe-SOD is localized in chloroplasts, which contain 80 to 90% of the cellular Fe pool (Kobayashi and Nishizawa 2012). Reasonably enhanced Fe accumulation in Mn-deficient plants suggests an upregulation of Fe-SOD in chloroplasts, which is a main site of O_2^- generation (Tewari et al. 2013a, b). Overall, Mn-deficient plants were shown to enhance total SOD and Mn-SOD activity (Tewari et al. 2013a, b; Yu and Rengel 1999). Enhanced SOD activity was found in Mn-deficient Norway spruce needles (Polle et al. 1992). Increased SOD activity may, consequently, be an index of O_2^- generation in Mn-deficient plants. Increased accumulation of H_2O_2 further substantiates production of O_2^- and oxidative damage (represented by increased lipid peroxidation) in Mn-deficient plants (Tewari et al. 2013a, b).

APX activity, a heme Fe-enzyme, has been found to be decreased in Mn-deficient plants (Tewari et al. 2013a, b; Yu et al. 1998). Also, decreased APX activity was reported in Mn-deficient Norway spruce (Polle et al. 1992). Increased ROS production in Mn-deficient mulberry is perhaps due to the loss of APX activity (Tewari et al. 2013a, b). APX is a high-affinity H_2O_2 scavenger (K_m 100 µM) compared to CAT (K_m 40 to 600 mM), which removes H_2O_2 at an accelerated rate (Mhamdi et al. 2010; Tewari et al. 2013a, b). Increased CAT activity, however, was futile in protecting Mn-deficient mulberry plants against oxidative stress. It is also reflected by higher lipid peroxidation and a disturbed redox environment of the cells, as measured by the DHA/

Table 5 Effect of manganese deficiency on different measures of oxidative metabolism in different plants and their parts

Parameter	Plant name	Plant part	Effect	Reference
Reactive oxygen species				
ROS	*Marchantia polymorpha*	Thallus	Increase	(Messant et al. 2022)
Superoxide anion (O_2^-)	*Morus alba*	Leaves	Increase	(Tewari et al. 2013a, b)
Hydrogen peroxide (H_2O_2)	*Capsicum annuum*	Young fruit	Increase	(Silber et al. 2009)
	Zea mays	Roots	Increase	(Zhao et al. 2014)
	Zea mays	Leaves	Increase	(Zhao et al. 2014)
	Sorghum bicolor	Leaves	Increase	(de Oliveira et al. 2019)
Oxidative damage				
Lipid peroxidation (LPX)	*Pinus sylvestris*	Needle	Increase	(Ivanov et al. 2022)
	Mentha piperita	Leaves	Increase	(Candan and Tarhan 2011)
	Morus alba	Leaves	Increase	(Tewari et al. 2013a, b)
	Sorghum bicolor	Leaves	Increase	(de Oliveira et al. 2019)
	Zea mays	Roots	Increase	(Zhao et al. 2014)
Ion leakage	*Hordeum vulgare*	Leaves	Increase	(Hebbern et al. 2009)
Antioxidants				
AA	*Picea abies*	Needle	Increase	(Polle et al. 1992)
	Phaseolus vulgaris	Leaves	Decrease	(Mehlhorn and Wenzel 1996)
	Mentha piperita	Leaves	Decrease	(Candan and Tarhan 2011)
	Morus alba	Leaves	Decrease	(Tewari et al. 2013a, b)
	Capsicum annuum	Young fruit	Increase	(Silber et al. 2009)
DHA	*Morus alba*	Leaves	NS	(Tewari et al. 2013a, b)
DHA/AA	*Morus alba*	Leaves	Increase	(Tewari et al. 2013a, b)
AA/DHA	*Zea mays*	Roots	Decrease	(Zhao et al. 2014)
GSH	*Picea abies*	Needle	NS	(Polle et al. 1992)
GSH/GSSG	*Zea mays*	Roots	Decrease	(Zhao et al. 2014)
Thiol/Disulfide	*Zea mays*	Roots	Decrease	(Zhao et al. 2014)
Enzyme activity				
SOD	*Picea abies*	Needles	Increase	(Polle et al. 1992)
	Mentha piperita	Leaves	Decrease	(Candan and Tarhan 2011)
	Morus alba	Leaves	Increase	(Tewari et al. 2013a, b)
	Capsicum annuum	Young fruit (apoplast)	Decrease	(Silber et al. 2009)
	Sorghum bicolor	Leaves	Decrease	(de Oliveira et al. 2019)
	Marchantia polymorpha	Thallus	Increase	(Messant et al. 2022)
	Zea mays	Roots	Decrease	(Zhao et al. 2014)
	Lycopersicon esculentum	Leaves	Increase	(Shenker et al. 2004)
	Nicotiana tabacum	Leaves	Decrease	(Yu et al. 1998, 1999)
	Lupinus angustifolius	Leaves	Increase	(Yu and Rengel 1999)
Mn-SOD	*Chlamydomonas reinhardtii*	Cells	Decrease	(Allen et al. 2007)
	Phaseolus vulgaris	Leaves	Increase	(Mehlhorn and Wenzel 1996)
	Lupinus angustifolius	Leaves	Increase	(Yu and Rengel 1999)
	Nicotiana tabacum	Leaves	Decrease	(Yu et al. 1998, 1999)
	Morus alba	Leaves	NS	(Tewari et al. 2013a, b)
	Lycopersicon esculentum	Leaves	Increase	(Shenker et al. 2004)
	Medicago truncatula	Leaves	Decrease	(López-Millán et al. 2005)
	Arabidopsis thaliana	Leaves	Decrease	(Lanquar et al. 2010)
	Picea abies	Needles	Decrease	(Kroniger et al. 1995)
	Phanerochaete chrysosporium	Mycelium	Decrease	(NATHAN ROTHSCHILD et al. 1999)
Fe-SOD	*Chlamydomonas reinhardtii*	Cells	Decrease	(Allen et al. 2007)
	Lupinus angustifolius	Leaves	Decrease	(Yu and Rengel 1999)

Table 5 (continued)

Parameter	Plant name	Plant part	Effect	Reference
	Nicotiana tabacum	Leaves	Decrease	(Yu et al. 1999)
	Morus alba	Leaves	Increase	(Tewari et al. 2013a, b)
Cu/Zn-SOD	*Lupinus angustifolius*	Leaves	Increase	(Yu and Rengel 1999)
	Nicotiana tabacum	Leaves	Increase	(Yu et al. 1999)
	Nicotiana tabacum	Leaves	Decrease	(Yu et al. 1998)
	Phaseolus vulgaris	Leaves	NS	(Mehlhorn and Wenzel 1996)
	Lycopersicon esculentum	Leaves	Increase	(Shenker et al. 2004)
	Morus alba	Leaves	NS*	(Tewari et al. 2013a, b)
	Medicago truncatula	Leaves	Increase	(López-Millán et al. 2005)
	Picea abies	Needles	Increase	(Kroniger et al. 1995)
CAT	*Mentha piperita*	Leaves	Decrease	(Candan and Tarhan 2011)
	Morus alba	Leaves	Increase	(Tewari et al. 2013a, b)
	Helianthus annuus	Leaves	Increase	(Weinstein and Robbins 1955)
	Vigna radiata	Leaves	Increase	(Sinha et al. 2006)
POD	*Morus alba*	Leaves	NS	(Tewari et al. 2013a, b)
	Vigna radiata	Leaves	Increase	(Sinha et al. 2006)
	Picea abies	Needles	Increase	(Polle and Chakrabarti 1994)
	Mentha piperita	Leaves	Decrease	(Candan and Tarhan 2011)
APX	*Picea abies*	Needle	NS	(Polle et al. 1992)
	Morus alba	Leaves	Decrease	(Tewari et al. 2013a, b)
	Nicotiana tabacum	Leaves	Decrease	(Yu et al. 1998)
	Sorghum bicolor	Leaves	Increase	(de Oliveira et al. 2019)
	Mentha piperita	Leaves	Increase	(Candan and Tarhan 2011)
	Zea mays	Roots	Decrease	(Zhao et al. 2014)
	Phaseolus vulgaris	Leaves	NS	(Mehlhorn and Wenzel 1996)
GR	*Picea abies*	Needle	NS	(Polle et al. 1992)
	Zea mays	Roots	Decrease	(Zhao et al. 2014)
Glutathione-S- transferase (GST)	*Zea mays*	Roots	Decrease	(Zhao et al. 2014)
MDHAR	*Picea abies*	Needle	Increase	(Polle et al. 1992)
Proteome				
FER1	*Arabidopsis thaliana*	Shoots	Decrease	(Lanquar et al. 2010)
Peroxidase AtPER44-like isoform 1 (AtPER44-like)	*Solanum lycopersicum*	Xylem sap	Decrease	(Ceballos-Laita et al. 2018)
Peroxidase AtPER12-like	*Solanum lycopersicum*	Xylem sap	Decrease	(Ceballos-Laita et al. 2018)
Cationic peroxidase 1 (AtPER52-like)	*Solanum lycopersicum*	Xylem sap	Decrease	(Ceballos-Laita et al. 2018)
Nectarin-1 (MnSOD-like)	*Solanum lycopersicum*	Xylem sap	Decrease	(Ceballos-Laita et al. 2018)
Mn-SOD (MSD1)	*Chlamydomonas reinhardtii*	Cells	Decrease	(Hsieh et al. 2013)
Mn-SOD (MSD3)	*Chlamydomonas reinhardtii*	Cells	Decrease	(Hsieh et al. 2013)
Transcripts				
Glutathione transferase (GSTU21)	*Arabidopsis thaliana*	Roots	Increase	(Wei Yang et al. 2008)
Glutathione transferase (GSTF5)	*Arabidopsis thaliana*	Roots	Increase	(Wei Yang et al. 2008)
Putative peroxidase (*At1g77100*)	*Arabidopsis thaliana*	Roots	Decrease	(Wei Yang et al. 2008)
Putative peroxidase (At2g24800)	*Arabidopsis thaliana*	Roots	Increase	(Wei Yang et al. 2008)
Mn-SOD (MSD3, MSD5)	*Chlamydomonas reinhardtii*	Cells	Increase	(Allen et al. 2007)
GPX3, GPX4	*Chlamydomonas reinhardtii*	Cells	Increase	(Allen et al. 2007)
APX1	*Chlamydomonas reinhardtii*	Cells	Increase	(Allen et al. 2007)
CAT1	*Chlamydomonas reinhardtii*	Cells	NS	(Allen et al. 2007)

NS means non-significant effect, *GSTs* encode glutathione S-transferase, *MSD3* and *MSD5* encode Mn-SODs

AA ratio. This high DHA/AA ratio may result from a larger decrease in AA and/or increased DHA content. Hence, a decrease in AA (an important antioxidant molecule) capable of removing ROS directly (Horemans et al. 2000), expected to expose Mn-deficient plants to oxidative damage more severely.

High POD activity has been revealed to be a characteristic of Mn-deficient plants (Polle and Chakrabarti 1994; Polle et al. 1992; Sinha et al. 2006). However, Mn-deficient mulberry did not increase POD activity. Enhanced Fe levels in Mn-deficient plants appear to be a reason for the increased POD activity, as PODs are Fe-dependent enzymes. The non-specific PODs oxidize many phenolic compounds, generating H_2O_2, hence implicated in lignification (Polle and Chakrabarti 1994). MDHAR and GR, which are respectively catalyze the reduction of monodehydroascorbate radical to AA and GSSG to GSH, have also been upregulated by Mn deficiency (Polle et al. 1992). Increased H_2O_2, DHA/AA ratio, along with upregulation in SOD (Tewari et al. 2013a, b), MDHAR, and GR (Polle et al. 1992), and downregulation in APX (Tewari et al. 2013a, b) suggest initiation of oxidative stress and limited working of the AA-GSH cycle in plants suffering from Mn deficiency. Mn deficiency leads to changes in ROS levels, and the antioxidant defense responses of different plants are summarized in Table 5. Moreover, changes in transcripts and proteome in Mn-deficient plants are also presented in Table 5.

Boron deficiency and oxidative stress

Boron (B) is an essential mineral element for growth and development in plants (Broadley et al. 2012) and as a borate it forms diester bonds with rhamnogalacturonan-II (RG-II) for cross-linking pectic polysaccharides in the cell walls (Begum et al. 2023; O'Neill et al. 2001). B uptake and transport in plants may be described through three main mechanisms: (i) passive diffusion occurring through the plasma membrane, (ii) facilitated diffusion occurring through water channels, viz., NIPs, nodulin 26-like intrinsic proteins, and (iii) active high-affinity transport coordinated by BOR transporters and get induced under low availability of B (González-Fontes et al. 2016; Kato et al. 2008; Takano et al. 2010). Levels of B in the earth's crust ranges between 20 and 200 μg g^{-1} and availability of B ranges between 0.4 and 5 μg g^{-1} of soils. Plant tissues B is generally measured to a level of 2 μmol g^{-1} (20 μg g^{-1}) of dry weight (Tewari et al. 2010). Uptake of B is by plants as non-dissociated H_3BO_3 (Broadley et al. 2012). At the cellular level, B had fundamental role in the sugar transport, formation of cell wall and their lignification, respiration, RNA, phenol, and IAA metabolisms (Broadley et al. 2012; Hänsch and Mendel 2009). Moreover, downregulation of various cell wall-related genes in the roots of *Arabidopsis* under B deficiency, shows the critical importance of B in cell wall formation (Camacho-Cristóbal et al. 2008). Inhibition in H^+ release upon B depletion, its recovery on B restoration to plants in the tissue culture medium, revealed B function in the cell membrane (Blevins and Lukaszewski 1998). B deficiency suppresses the genes encoding histones and chromatin assembly and remodelling and impairs the activity of shoot apical meristem (Chen et al. 2024). It also affects root elongation, lateral root initiation, root hair formation and geotropism (Chen et al. 2023).

B is comparatively immobile, hence its deficiency symptoms, which first develop in young and developing organs. B-deficient plants exhibit abnormal or retarded growth in their shoot apices. Under severe B deficiency, the stems and leaves of plants become inelastic, which indicates the function of B in plant-water relations (Tewari et al. 2010). Mulberry plants with B deficiency exhibit cracked stems and petioles, develop scaly surfaces, and form internal and external corky structures (Tewari et al. 2010). B deficiency reduces transpiration due to poor differentiation and collapse of vascular tissues with low stomatal density (Panter et al. 2023). Excessive ROS generation probably promotes B deficiency symptoms in the stem and petioles, as represented by necrosis and cracks (Tewari et al. 2010). The occurrence of an oxidizing environment, due to higher ROS production (Michael and Krishnaswamy 2012; Tewari et al. 2010), causes senescence or necrosis in B-deficient plants (Schafer and Buettner 2001). The enzymes associated with induction of senescence and necrosis, viz., RNase, acid phosphatase, and polyphenol oxidase (Kar and Mishra 1976; Veljovic-Jovanovic 2006), have been elevated in B-deficient plants (Pandey and Archana 2011; Sinha et al. 2009). The buildup of various phenolics and an increase in the activity of polyphenol oxidase in B-deficient plants have also been shown to produce reactive derivatives of phenolics and ROS (Broadley et al. 2012; Dordas and Brown 2005).

Enhanced buildup of ROS and lipid peroxides, along with a fall in the AA pool in B-depleted BY-2 cells of tobacco exhibiting cell death, have been observed by Koshiba et al. (2009). MDA (lipid peroxidation) accumulation in B-deficient plants is an index of oxidative damage (Hajiboland and Bastani 2012; Michael and Krishnaswamy 2012; Pandey and Archana 2013; Tewari et al. 2010). Leaves of B-deficient plants have been shown to accumulate excess Fe (Tewari et al. 2010), which is anticipated to be important for $OH^{\cdot}$ production and lipid peroxidation (Stohs and Bagchi 1995). B-deficient plants produce excess O_2^- by enhancing NADPH oxidase activity (Camacho-Cristóbal et al. 2015). The O_2^- (Mukhopadhyay et al. 2013) and H_2O_2 (Mukhopadhyay et al. 2013; Tewari et al. 2010) accumulations also cause an AA decrease and an increase in the DHA/AA ratio in the B-deficient mulberry leaves

(Tewari et al. 2010). Similarly, a decline in AA concentration was observed by Koshiba et al. (2009) in the B-deficient BY-2 cells of tobacco. Increased DHA/AA suggests the operation of the AA-GSH cycle in B-deficient plants. A decrease in AA (an antioxidant molecule), which has the capability to directly quench ROS (Horemans et al. 2000), is likely to induce oxidative challenges in B-deficient plants.

Increased accumulations of ROS activate enzymes responsible for removing ROS in B-deficient plants. B-deficient plants exhibit higher activity of SOD along with the expression of additional SOD isoforms in the mulberry leaves (Tewari et al. 2010). Enhanced SOD activity has been observed in tea (Hajiboland and Bastani 2012; Mukhopadhyay et al. 2013), citrus (Han et al. 2008), cowpea (Michael and Krishnaswamy 2012), maize (Pandey and Archana 2013), and bitter gourd (Sinha et al. 2009) plants exposed to B-deficient conditions, which suggests that B deficiency imposes oxidative stress in these plants. Thus, H_2O_2 generated due to enhanced SOD activity is removed by H_2O_2 scavenging enzymes. Increases in the APX along with the DHA/AA ratio in B-deficient plants indicate activation of the AA-GSH cycle (Tewari et al. 2010). H_2O_2 scavenging enzymes like CAT and POD are likewise enhanced by B deficiency (Tewari et al. 2010). Increased antioxidant enzyme activity on a protein basis has been described in several other plant species, viz., tea (Hajiboland and Bastani 2012; Hajiboland et al. 2011; Mukhopadhyay et al. 2013), cow pea (Michael and Krishnaswamy 2012), bittergourd (Sinha et al. 2009), and maize (Pandey and Archana 2013), subjected to B deficiency. Also, increases in GR activity have been recorded in B-deficient citrus (Han et al. 2008) and maize (Pandey and Archana 2013). Enhanced GR activity suggests operation of the AA-GSH cycle to reduce GSSG to GSH for effective operation of the pathway. Other antioxidant enzymes of the AA-GSH cycle, viz., MDHAR and DHAR, responsible for the MDHA and DHA reduction into AA, are upregulated and downregulated, respectively, in citrus leaves suffering from B-deficiency (Han et al. 2008). On the contrary, Shah et al. (2017) observed decreased SOD, POD, CAT, and APX activity on a fresh weight basis in B-deficient citrange plants.

Under B deficiency, tea plantlets also upregulated the expressions of CAT, APX, and SOD genes (Mukhopadhyay et al. 2013), which corroborates the increased activity of the above-said enzymes in tea as well as other plant species. Koshiba et al. (2009) also observed an upregulation of the *APX* genes of tobacco in B-deficient BY-2 cell suspension cultures. Supplying lipophilic antioxidants efficiently blocked the death of BY-2 cells in tobacco, which is B-deprived. These results propose that oxidative damage is the main reason for cell death in B-deficient BY-2 cells. Moreover, this study states that B deprivation causes induction in an increase in $[Ca^{2+}]_{cyt}$ in roots of *A. thaliana* (Quiles-Pando et al. 2019). This B deprivation-mediated increase in $[Ca^{2+}]_{cyt}$ is largely due to the apoplastic Ca^{2+} influx through the plasma membrane-located Ca^{2+} channels in an IP_3-dependent manner (Quiles-Pando et al. 2019). The rise in $[Ca^{2+}]_{cyt}$ is observed to be mediated by oxidative bursts and at onset of PCD (Zuppini et al. 2004). However, BY2 cells did not exhibit typical signs of PCD, viz., internucleosomal DNA fragmentation, declined *cAPX* expression, and cycloheximide-mediated defense in cell death (Koshiba et al. 2009). High H_2O_2 concentrations along with high Fe concentrations in the B-deficient leaves may activate the Fenton or Haber–Weiss reaction, leading to $^{\cdot}OH$ radical generation and lipid peroxidation in the B-deficient leaves. Increased antioxidant enzyme activity has probably reduced oxidative stress. B deficiency-induced alterations in ROS and its antioxidant defense responses in different plants are presented in Table 6. Moreover, changes in transcripts and proteome in B-deficient plants are presented in Table 6.

Molybdenum deficiency and oxidative stress

Molybdenum (Mo) is a micronutrient for organisms that is present in small amounts in the soil (ranging from 0.2 to 36 mg kg^{-1}). Mo occurs primarily as the molybdate oxyanion (MoO_4^{2-}) in aqueous solutions with a pH > 4.3, in its highest oxidized state [Mo (VI)]. While at a lower pH (< 4.3), protonated species like $HMoO_4^-$, and $MoO_3(H_2O)_3$ became the prevailing forms (Broadley et al. 2012). Mo concentrations may vary from plant to plant. In leaf tissue, it may vary between 0.05 and 2.0 µg g^{-1} of dry weight (Römheld 2012). Visually, the symptoms of Mo deficiency in plants are closely related to those of N deficiency. Symptoms of Mo deficiency include stunted growth, leaf size reduction, general chlorosis, distortion or cupping of middle or old leaves, marginal scorching, and necrosis in the leaf tip (Chatterjee et al. 1985; Kaiser et al. 2005). A deficiency in Mo affects flower maturation and fruiting. The reason is the conspicuous effect of the Mo unavailability for pollen and anther development or the failures of anthesis under Mo deficiency as reported previously (Agarwala et al. 1979; Gopal and Shukla 2016).

Plant requirements for Mo are moderately less than any of the other mineral elements. Mo is mobile element both in xylem and phloem (Broadley et al. 2012). There are few Mo-containing proteins in plants (Hänsch and Mendel 2009). These proteins participate in N assimilation, metabolism of S, biosynthesis of hormones, and stress-related reactions (Hänsch and Mendel 2009). Mo-containing enzymes: nitrate reductase (catalyzes the first reaction in nitrate assimilation, which is the reduction of nitrate into nitrite), nitrogenase (found in nitrogen fixing bacteria and plays a crucial function in biological nitrogen fixation), aldehyde oxidase (catalyses the last step of the biosynthesis of ABA), and sulphite oxidase (protects the plant against noxious concentrations

Table 6 Effect of boron deficiency on different measures of oxidative metabolism in different plants and their parts

Parameter	Plant name	Plant part	Effect	Reference
ROS				
Superoxide anion (O_2^-)	*Camellia sinensis*	Leaves	Increase	(Mukhopadhyay et al. 2013)
	Camellia sinensis	Roots	Increase	(Mukhopadhyay et al. 2013)
	Brassica napus	Roots	Increase	(Zhou et al. 2017)
	Brassica napus	Cells	Increase	(Zhou et al. 2017)
	Arabidopsis thaliana	Roots	Increase	(Camacho-Cristóbal et al. 2015)
	Citrus sinensis	Leaves	Increase	(Han et al. 2008)
Hydrogen peroxide (H_2O_2)	*Camellia sinensis*	Roots	NS	(Hajiboland and Bastani 2012; Hajiboland et al. 2011)
	Camellia sinensis	Roots	Increase	(Mukhopadhyay et al. 2013)
	Camellia sinensis	Leaves	Increase	(Mukhopadhyay et al. 2013)
	Morus alba	Leaves	Increase	(Tewari et al. 2010)
	Rosa damascena	Cells	Increase	(Dordas and Brown 2005)
	Brassica napus	Roots	Increase	(Zhou et al. 2017)
	Brassica napus	Cells	Increase	(Zhou et al. 2017)
	Brassica rapa	Leaves	Increase	(Hajiboland and Farhanghi 2010)
	Brassica rapa	Roots	Increase	(Hajiboland and Farhanghi 2010)
	Vigna unguiculata	Leaves	Increase	(Michael and Krishnaswamy 2012)
	Zea mays	Leaves	Increase	(Pandey and Archana 2013)
Oxidative damage				
Electrolyte leakage	*Brassica napus*	Roots	Increase	(Zhou et al. 2017)
	Citrus sinensis	Leaves	Increase	(Han et al. 2008)
	Helianthus annuus	Seedling	Increase	(Cakmak and Romheld 1997)
	Vigna unguiculata	Leaves	Increase	(Michael and Krishnaswamy 2012)
Lipid peroxidation	*Camellia sinensis*	Leaves	Increase	(Hajiboland and Bastani 2012; Hajiboland et al. 2011; Mukhopadhyay et al. 2013)
	Camellia sinensis	Roots	Increase	(Hajiboland and Bastani 2012; Hajiboland et al. 2011; Mukhopadhyay et al. 2013)
	Zea mays	Leaves	Increase	(Pandey and Archana 2013)
	Vigna unguiculata	Leaves	Increase	(Michael and Krishnaswamy 2012)
	Citrus sinensis	Leaves	Increase	(Han et al. 2008)
	Brassica napus	Roots	Increase	(Zhou et al. 2017)
	Citrange	Leaves	Decrease	(Shah et al. 2017)
	Momordica charantia	Leaves	Increase	(Sinha et al. 2009)
	Morus alba	Leaves	NS	(Tewari et al. 2010)
	Nicotiana tabacum	Cells	Increase	(Koshiba et al. 2009)
Antioxidants				
AA	*Morus alba*	Leaves	Decrease	(Tewari et al. 2010)
	Citrus sinensis	Leaves	Increase	(Han et al. 2008)
	Helianthus annuus	Leaves	Decrease	(Cakmak and Romheld 1997)
	Zea mays	Leaves	Decrease	(Pandey and Archana 2013)
	Nicotiana tabacum	Cells	Decrease	(Koshiba et al. 2009)
	Vigna unguiculata	Leaves	Decrease	(Michael and Krishnaswamy 2012)
DHA	*Morus alba*	Leaves	Decrease	(Tewari et al. 2010)
	Helianthus annuus	Leaves	Increase	(Cakmak and Romheld 1997)
	Zea mays	Leaves	Decrease	(Pandey and Archana 2013)
DHA/AA	*Morus alba*	Leaves	NS	(Tewari et al. 2010)
	Helianthus annuus	Leaves	Increase	(Cakmak and Romheld 1997)
GSH	*Citrus sinensis*	Leaves	Decrease	(Han et al. 2008)
	Nicotiana tabacum	Cells	Increase	(Koshiba et al. 2009)

Table 6 (continued)

Parameter	Plant name	Plant part	Effect	Reference
Non-protein Thiol	*Helianthus annuus*	Leaves	Decrease	(Cakmak and Romheld 1997)
	Zea mays	Leaves	Increase	(Pandey and Archana 2013)
Enzyme activity				
NADPH oxidase	*Arabidopsis thaliana*	Roots	Increased	(Camacho-Cristóbal et al. 2015)
SOD	*Camellia sinensis*	Leaves	Increase	(Hajiboland et al. 2011; Mukhopadhyay et al. 2013)
	Camellia sinensis	Roots	Increase	(Hajiboland and Bastani 2012; Hajiboland et al. 2011; Mukhopadhyay et al. 2013)
	Camellia sinensis	Leaves	NS	(Hajiboland and Bastani 2012)
	Brassica rapa	Leaves	Increase	(Hajiboland and Farhanghi 2010)
	Brassica rapa	Roots	Increase	(Hajiboland and Farhanghi 2010)
	Vigna unguiculata	Leaves	Increase	(Michael and Krishnaswamy 2012)
	Morus alba	Leaves	Increase	(Tewari et al. 2010)
	Citrus sinensis	Leaves	Increase	(Han et al. 2008)
	Citrange orange	Leaves	Decrease	(Shah et al. 2017)
	Helianthus annuus	Leaves	NS	(Cakmak and Romheld 1997)
	Momordica charantia	Leaves	Increase	(Sinha et al. 2009)
	Nicotiana tabacum	Cells	Increase	(Koshiba et al. 2009)
	Zea mays	Leaves	Increase	(Pandey and Archana 2013)
	Medicago sativa	Flower/Seed	Decrease	(Chen et al. 2021)
CAT	*Morus alba*	Leaves	Increase	(Tewari et al. 2010)
	Zea mays	Pollen	Decrease	(Agarwala et al. 1981)
	Zea mays	Leaves	Increase	(Pandey and Archana 2013)
	Citrus sinensis	Leaves	Decrease	(Han et al. 2008)
	Citrange orange	Leaves	Decrease	(Shah et al. 2017)
	Brassica rapa	Leaves	NS	(Hajiboland and Farhanghi 2010)
	Brassica rapa	Roots	NS	(Hajiboland and Farhanghi 2010)
	Camellia sinensis	Leaves	Increase	(Mukhopadhyay et al. 2013)
	Camellia sinensis	Roots	Increase	(Mukhopadhyay et al. 2013)
	Nicotiana tabacum	Cells	NS	(Koshiba et al. 2009)
	Momordica charantia	Leaves	Decrease	(Sinha et al. 2009)
	Brassica napus	Roots	Increase	(Zhou et al. 2017)
	Medicago sativa	Flower/Seed	Decrease	(Chen et al. 2021)
POD	*Morus alba*	Leaves	Increase	(Tewari et al. 2010)
	Zea mays	Pollen	Decrease	(Agarwala et al. 1981)
	Zea mays	Leaves	Increase	(Pandey and Archana 2013)
	Citrange orange	Leaves	Decrease	(Shah et al. 2017)
	Brassica rapa	Leaves	Increase	(Hajiboland and Farhanghi 2010)
	Brassica rapa	Roots	Increase	(Hajiboland and Farhanghi 2010)
	Brassica napus	Roots	Increase	(Zhou et al. 2017)
	Camellia sinensis	Leaves	Increase	(Hajiboland and Bastani 2012; Mukhopadhyay et al. 2013)
	Camellia sinensis	Roots	Increase	(Hajiboland and Bastani 2012; Mukhopadhyay et al. 2013)
	Vigna unguiculata	Leaves	Increase	(Michael and Krishnaswamy 2012)
	Momordica charantia	Leaves	Increase	(Sinha et al. 2009)
	Medicago sativa	Flower/Seed	Decrease	(Chen et al. 2021)
APX	*Morus alba*	Leaves	Increase	(Tewari et al. 2010)
	Zea mays	Leaves	Increase	(Pandey and Archana 2013)
	Citrange orange	Leaves	Decrease	(Shah et al. 2017)
	Brassica rapa	Leaves	Increase	(Hajiboland and Farhanghi 2010)

Table 6 (continued)

Parameter	Plant name	Plant part	Effect	Reference
	Brassica rapa	Roots	Increase	(Hajiboland and Farhanghi 2010)
	Camellia sinensis	Leaves	Increase	(Hajiboland and Bastani 2012; Mukhopadhyay et al. 2013)
	Camellia sinensis	Roots	Increase	(Hajiboland and Bastani 2012; Mukhopadhyay et al. 2013)
	Citrus sinensis	Leaves	Increase	(Han et al. 2008)
	Helianthus annuus	Leaves	Increase	(Cakmak and Romheld 1997)
	Nicotiana tabacum	Cells	Increase	(Koshiba et al. 2009)
	Momordica charantia	Leaves	Increase	(Sinha et al. 2009)
MDHAR	*Citrus sinensis*	Leaves	Increase	(Han et al. 2008)
DHAR	*Citrus sinensis*	Leaves	Decrease	(Han et al. 2008)
GR	*Citrus sinensis*	Leaves	Increase	(Han et al. 2008)
	Zea mays	Leaves	Increase	(Pandey and Archana 2013)
Proteome				
Pathogenesis related protein	*Medicago sativa*	Flower	Increase	(Chen et al. 2021)
Late embryogenesis abundant (LEA)	*Medicago sativa*	Seed	Increase	(Chen et al. 2021)
Heat shock protein I	*Medicago sativa*	Seed	Decrease	(Chen et al. 2021)
Cu/Zn-SOD	*Arabidopsis thaliana*	Leaves	Decrease	(Chen et al. 2014)
Peroxiredoxin 2E	*Arabidopsis thaliana*	Leaves	Decrease	(Chen et al. 2014)
Glutathione S-transferase (GST)	*Arabidopsis thaliana*	Leaves	NS	(Chen et al. 2014)
Transcripts				
RBOHDs	*Brassica napus*	Roots	Increase	(Zhou et al. 2017)
RBOHFs	*Brassica napus*	Roots	Increase	(Zhou et al. 2017)
SOD	*Camellia sinensis*	Leaves	Increase	(Mukhopadhyay et al. 2013)
	Camellia sinensis	Roots	Increase	(Mukhopadhyay et al. 2013)
cAPX	*Nicotiana tabacum*	Cells	Increase	(Koshiba et al. 2009)
APX	*Camellia sinensis*	Leaves	Increase	(Mukhopadhyay et al. 2013)
	Camellia sinensis	Roots	Increase	(Mukhopadhyay et al. 2013)
CAT	*Camellia sinensis*	Leaves	NS	(Mukhopadhyay et al. 2013)
	Camellia sinensis	Roots	NS	(Mukhopadhyay et al. 2013)
POD	*Citrus sinensis*	Leaves	Increase	(Liu et al. 2017)
	Populus × *euramericana*	Roots	Increase	(Su et al. 2019)

NS means non-significant effect *RBOHs* encode NADPH oxidases, *cAPX* encode cytosolic ascorbate peroxidase

of sulphite) (Hänsch and Mendel 2009; Kaiser et al. 2005). Thus Mo-deficient plant shows low activity of NR, leading to accumulation of nitrate, and suppression in nitrite, ammonium, amino acid and protein levels (Cakmak et al. 2023).

Catalytically, Mo appears to be inactive in biological systems until it gets complexed by a special cofactor (Hänsch and Mendel 2009). Mo is bound to a Mo cofactor (MoCo), a pterin compound that binds to various apoproteins (Mendel and Hänsch 2002). Enzymes participating in MoCo biosynthesis contain Fe-S clusters. Certain Mo-containing enzymes produce ROS (aldehyde oxidase, AO; sulfite oxidase, SO; xanthine dehydrogenase, XDH) or NO (nitrate reductase, NR) as side reaction products (Hänsch and Mendel 2009). *Arabidopsis* ABA3, an enzyme involved in the production of the sulfurated form of MoCo, which is necessary for the AO and XDH enzymatic activity (Watanabe et al. 2018). Watanabe et al. (2018) presented that the anthocyanin increase due to oxidative stress was normally observed in wild-type plants and was diminished in mutant plants of *aba3*. Thus, under stress, *aba3* mutants build up less ABA and anthocyanin but exhibit more electrolyte leakage, which is indicative of membrane damage (Watanabe et al. 2018). AO and XDH may play a role in the generation of NO and ROS in plants (Wu et al. 2017). Being Mo-requiring enzymes, AO and XDH activities are expected to decrease under Mo deficiency, resulting in relatively low ROS formation too. However, under drought stress, Mo-deficient wheat plants accumulate a greater amount of O_2^-, and H_2O_2 and consequently cause more lipid peroxidation in their leaves (Wu et al. 2017). Stimulation in certain antioxidant enzymes, like CAT, SOD and POD, activities were seen in chickpea plants grown at low Mo supply (Gopal and Shukla 2016). Increased POD activity has also been observed in Mo-deficient maize leaves;

Table 7 Effect of molybdenum deficiency on different measures of oxidative metabolism in different plants and their parts

Parameter	Plant name	Plant part	Effect	Reference
Reactive oxygen/nitrogen species				
Superoxide anion	*Triticum aestivum*	Leaves	Increase	(Wu et al. 2017)
Hydrogen peroxide	*Triticum aestivum*	Leaves	Increase	(Wu et al. 2017)
Nitric oxide	*Triticum aestivum*	Leaves	Decrease	(Wu et al. 2017)
Oxidative damage				
K^+ leakage	*Triticum aestivum*	Leaves	Increase	(Li et al. 2001)
Lipid peroxidation (LPX)	*Cicer arientinum*	Leaves	Decrease	(Gopal and Shukla 2016)
	Triticum aestivum	Leaves	Increase	(Wu et al. 2017)
Enzyme activity				
SOD	*Cicer arientinum*	Leaves	Increase	(Gopal and Shukla 2016)
	Triticum aestivum	Leaves	Decrease	(Wu et al. 2017)
	Triticum aestivum	Leaves	Decrease	(Li et al. 2001)
CAT	*Zea mays*	Leaves	Decrease	(Agarwala et al. 1978)
	Zea mays	Pollen	Increase	(Agarwala et al. 1979)
	Cicer arientinum	Leaves	Increase	(Gopal and Shukla 2016)
	Triticum aestivum	Leaves	Decrease	(Apostolova and Yaneva 2006)
	Triticum aestivum	Leaves	Decrease	(Wu et al. 2017)
	Triticum aestivum	Leaves	NS*	(Li et al. 2001)
POD	*Zea mays*	Leaves	Increase	(Agarwala et al. 1978)
	Zea mays	Pollen	Increase	(Agarwala et al. 1979)
	Cicer arientinum	Leaves	Increase	(Gopal and Shukla 2016)
	Triticum aestivum	Leaves	NS*	(Apostolova and Yaneva 2006)
	Triticum aestivum	Leaves	Decrease	(Wu et al. 2017)
	Triticum aestivum	Leaves	Increase	(Li et al. 2001)
APX	*Triticum aestivum*	Leaves	Increase	(Apostolova and Yaneva 2006)
	Triticum aestivum	Leaves	Decrease	(Wu et al. 2017)
GR	*Triticum aestivum*	Leaves	Increase	(Apostolova and Yaneva 2006)
Nitrate reductase	*Cicer arientinum*	Leaves	Decrease	(Gopal and Shukla 2016)
	Brassica campestris	Leaves	Decrease	(Chatterjee et al. 1985)
	Triticum aestivum	Leaves	Decrease	(Wu et al. 2017)
Proteome				
Rieske iron–sulfur protein	*Triticum aestivum*	Leaves	Increase	(Sun et al. 2014)
Xanthine dehydrogenase (XDH)	*Arabidopsis thaliana*	Shoot	Decrease	(Teschner et al. 2010)
Aldehyde oxidase (AO)	*Arabidopsis thaliana*	Shoot	Decrease	(Teschner et al. 2010)
Nitrate reductase (NR)	*Arabidopsis thaliana*	Shoot	Decrease	(Teschner et al. 2010)
Transcripts				
SOD	*Triticum aestivum*	Leaves	Decrease	(Wu et al. 2017)
CAT	*Triticum aestivum*	Leaves	Decrease	(Wu et al. 2017)
APX	*Triticum aestivum*	Leaves	Decrease	(Wu et al. 2017)
Nitrate reductase (NR1)	*Arabidopsis thaliana*	Shoot	Increase	(Ide et al. 2011)
Nitrate reductase (NR2)	*Arabidopsis thaliana*	Shoot	Increase	(Ide et al. 2011)
Nitrite reductase (NiR)	*Arabidopsis thaliana*	Shoot	Increase	(Ide et al. 2011)
Peroxidase (PER22)	*Arabidopsis thaliana*	Shoot	Increase	(Ide et al. 2011)
Peroxidase, putative	*Arabidopsis thaliana*	Shoot	Increase	(Ide et al. 2011)
Pathogenesis related protein (PRB1)	*Arabidopsis thaliana*	Shoot	Increase	(Ide et al. 2011)
Ferredoxin:NADP(H) oxidoreductase (RFNR1)	*Arabidopsis thaliana*	Shoot	Increase	(Ide et al. 2011)
Glutaredoxin family protein	*Arabidopsis thaliana*	Shoot	Increase	(Ide et al. 2011)
Heat shock factor (HSF4)	*Arabidopsis thaliana*	Shoot	Increase	(Ide et al. 2011)
Heat shock transcription factor (HSFC1)	*Arabidopsis thaliana*	Shoot	Increase	(Ide et al. 2011)
Heat shock protein-related	*Arabidopsis thaliana*	Shoot	Increase	(Ide et al. 2011)

Table 7 (continued)

Parameter	Plant name	Plant part	Effect	Reference
Glutaredoxin (GRX480)	*Arabidopsis thaliana*	Shoot	Increase	(Ide et al. 2011)

NS means non-significant effect, *RBOHs* encode NADPH oxidases, *cAPX* encode cytosolic ascorbate peroxidase

however, they decreased CAT activity (Agarwala et al. 1978). Pollen grains of Mo-deficient maize show enhanced CAT and POD activities (Agarwala et al. 1979). Enhanced SOD activity resulted from enhanced O_2^- accumulation in Mo-deficient plants. Mo deficiency-mediated ROS generation is probably manifested by upregulation in the POD, SOD, and CAT activities to deal with excess ROS produced in Mo-deficient plants (Gopal and Shukla 2016). Mo deficiency-induced modifications in ROS levels and their antioxidant defense regulation in various plants are presented in Table 7. Moreover, changes in transcripts and proteome in Mo-deficient plants are also presented in Table 7.

Nickel deficiency may induce oxidative stress

Micronutrients such as nickel and chlorine are vital for the proper growth and physiology of plants. Chemically, nickel (Ni) relates to Fe and Co. The preferable oxidation state of Ni is Ni^{2+} (Ni II) in biological systems. The Ni concentration in plants ranges from 0.05 to 5.0 µg g^{-1} dry mass (Broadley et al. 2012). Ni forms stable complexes with cysteine, histidine, and citrate and gets coordinated to various ligands in the case of Ni-containing enzymes. Important Ni-associated enzymes are Ni-SOD, CO dehydrogenase, Ni-dependent glyoxylase, acetyl-CoA decarbonylase synthase, and methyleneurease (Broadley et al. 2012). Plants have both Zn-type and Ni-type glyoxalases (GLY I) enzyme (Balparda et al. 2023). Urease and Ni-urease accessory protein Eu3 had an important role in plants (Broadley et al. 2012). Ni-SOD is quite like Fe-SOD and Mn-SOD. It catalyzes the disproportionation of O_2^- by redox cycling between the oxidation states of M(III) and M(II) (Ryan et al. 2010). Ni does not catalyze disproportionation of O_2^- in a neutral aqueous solution, probably due to lack of an attainable one-electron redox process (Ryan et al. 2010). Being a constituent of Ni-SOD, Ni participates in the adjustment of oxidative stress in plants. However, information associated with Ni deficiency-induced oxidative stress and its antioxidant properties and responses in plants is largely unknown.

Chlorine deficiency may induce oxidative stress

Chlorine is ubiquitous in plants. It occurs in the monovalent anionic form, chloride (Cl^-) in aqueous solutions. In plant cells, Cl^- is one of the abundant inorganic anions. Cl deficiency observed in high precipitation areas where sandy soils occur or can be created in artificial experiments proving its requirement in higher plants (Hänsch and Mendel 2009). Typically, Cl deficiency symptoms include wilting of leaves and leaf margins. Young leaves get curled, followed by shriveling and necrosis under severe deficiency (Broadley et al. 2012). Cl occurs in a free anion form or is bound loosely to ion exchange sites in plants. Cl^- functions as a counter anion for stabilizing the membrane potential and participates in pH and turgor regulation (Broadley et al. 2012). It has a central role in the photosynthetic oxygen-evolving complex (OEC). Onion (*Allium cepa*), does not synthesize malate in the guard cells; in such plants, Cl^- plays an important counter-ion role for the functioning of stomata, and its opening gets inhibited in the Cl-deficient plants. The opening and closing of the guard cells are regulated by the flux exchange of K^+ ions and anions like malate and chloride (Hänsch and Mendel 2009). Proton-pumping ATPase, and V-type ATPase located on the endomembrane system were stimulated by Cl^- (Broadley et al. 2012). A deficiency of Cl may suppress stomatal opening, resulting in the accumulation of reducing power and the consequent generation of ROS due to electrons spillage.

Opinion on ROS signaling under micronutrient deficiency

High rates of electron transfer in an oxygenic cellular environment are directly related to the production of ROS by electron leakage to O_2 (Waszczak et al. 2018). ROS-scavenging SOD is present in all life forms, suggesting the ROS generation in the early evolution of life on earth (Mittler 2017). The ROS signaling network appears to be well conserved among aerobic organisms, and they might use ROS for sensing nearby or invading cells during the course of evolution (Mittler et al. 2011). The signaling of ROS involves the perception, propagation, and ROS transducing signals from a cell to the next cell and beyond (Mittler 2017; Mittler et al. 2011). ROS has the property to propagate itself

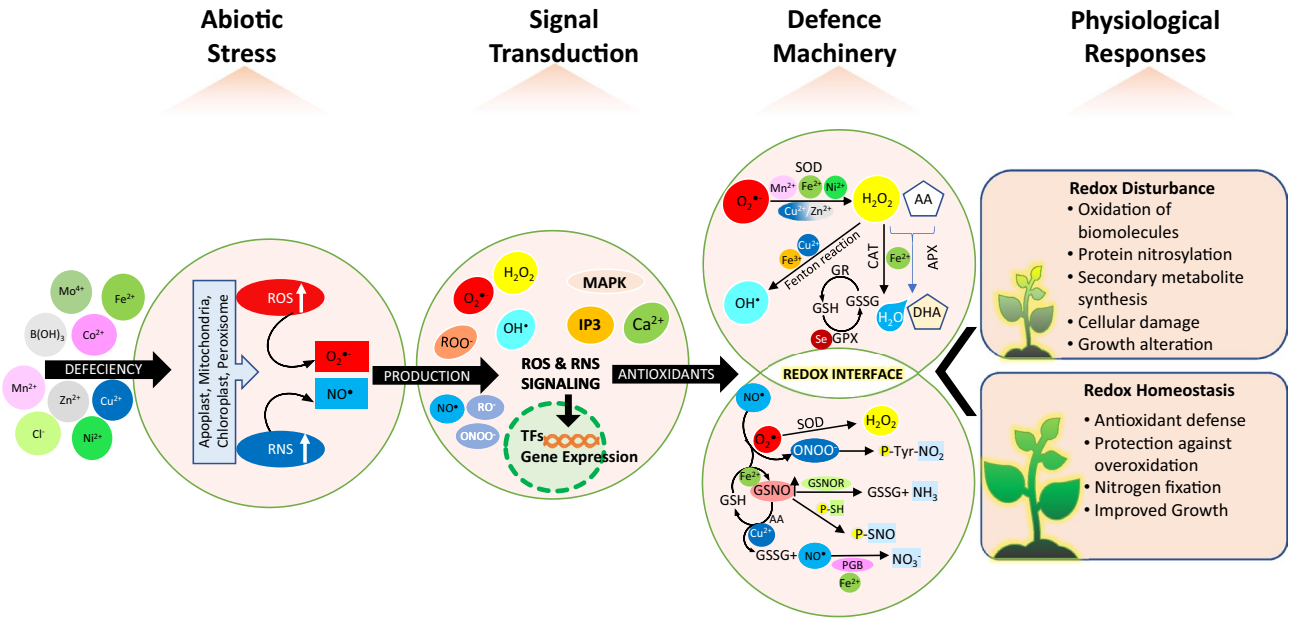


Fig. 5 Schematic representation of micronutrient deficiency-induced ROS and RNS coordination and regulation in plants initiating from production (most left circle) to activation of antioxidant/nitro-oxidative stress machinery and ending to physiological response (box 1: homeostasis and box 2: disturbances) under controlled signaling cascades to reduce the stress generated in the plant system. The ROS and RNS metabolism has been coordinated by multiple messengers viz, Ca^{2+}, inositol trisphosphate (IP3), and MAPKs (mitogen activated protein kinase), which probably phosphorylate related transcriptional factors (TFs) [middle circle], probably responsible for expression of catalase (CAT), ascorbate peroxidases (APX), superoxide dismutases (SOD), peroxidases (POD), S-nitrosoglutathione reductase (GSNOR) and biosynthesis of ascorbic acid, (AA), glutathione (GSH), *S*-nitrosoglutathione (GSNO), protein *S*-nitrosylation (P-SNO) [most right circle]. RNS-homeostasis involves the action of GSNOR, P-SH (protein thiol), PGB (phytoglobin) and AA (ascorbic acid) respectively in the reduction of GSNO into GSSG and NH_3, *S*-nitrosylation of proteins (P-SNO), conversion of NO into NO_3^-, GSH into GSSG and NO

throughout the plant body rapidly and efficiently. A differentiated cell has the individual capacity to activate the ROS signaling pathway in an autonomous manner (Mittler et al. 2011). Non-toxic levels need to be maintained and coordinated between ROS-producing and ROS-scavenging processes for utilizing ROS as a signaling molecule (Mittler et al. 2004). ROS regulates various biological functions related to growth, development, and their responses under stress (Mittler et al. 2011). Changes in RNS and ROS concentrations in cells may alter the forms and functions of several proteins as depicted in Fig. 5. This ROS signaling process is coordinated by oxidative post-translational modifications (oxidative-PTMs) (Mittler et al. 2022). ROS-mediated changes in the ratio of GSH/GSSG can cause oxidative-PTMs of cysteine residues. The oxidative-PTMs may cause conformation shifts in proteins [for example, in kinases [MITOGEN-ACTIVATED PROTEIN KINASE (MAPK) cascades, includes: MEKK1-MM1/2-MPK4/6 cascade, and Ser/Thr kinases], phosphatases [CLASS 2 PROTEIN PHOSPHATASE, PP2A or PP2C, TYR PHOSPHATASE, PTP], and transcription factors [NON-EXPRESSOR OF PATHOGENESIS-RELATED GENES 1, NPR1; C-REPEAT BINDING FACTORs, CBFs; HEAT SHOCK TRANSCRIPTION FACTORs, HSFs; ANAC089, MYB30, and RADICAL-INDUCED CELL DEATH 1, RCD1] (Mittler et al. 2022). ROS can modify the properties and functions (activity, specificity, and location) of these proteins, leading to the activation or suppression of stress-related responsive signal transduction paths (Mittler et al. 2022). ROS-mediated shifts in the ratio of GSH/GSSG might induce oxidative PTMs of cysteine residues of various proteins, including receptors, propagation and signal transducers, various transcription factors, and RBOHs (respiratory burst oxidase homologues or NADPH oxidase) through S-glutathionylation (Suzuki et al. 2011). Moreover, ROS-sensing is also coordinated by ROS and redox-sensitive proteins or oxidatively damaged metabolites such as β-carotene (Waszczak et al. 2018).

In reaction to stress stimuli, early signaling events comprising Ca^{2+} release into the cytosolic environment (Waszczak et al. 2018), activation of the MAPKs pathway, and protein phosphorylation are generally observed (Benschop et al. 2007; Steinhorst and Kudla 2013). Also, the process of ROS signaling is quite rapid and may be activated in a few seconds or minutes (Benschop et al. 2007). The contribution of ROS to environmental challenges may initiate with early bursts of oxidative responses leading to

ROS generation and the initialization of various signaling processes (Mittler et al. 2011; Nishimura and Dangl 2010). Additionally, the differential expressions of variously targeted RBOHs in several tissues and in different cells take place, indicating high signal specificity and the spatial coordination of ROS signaling processes (Suzuki et al. 2011).

Superoxide anion (O_2^-) is generally produced in the apoplast by cell membrane through the catalytic action of RBOHs (Dubiella et al. 2013). The RBOH is a cell membrane-integrated protein (Tewari et al. 2021a, b). The dismutation of O_2^- to H_2O_2 occurs spontaneously or by the catalytic action of SODs (Mittler et al. 2011, 2022). H_2O_2 is a membrane-permeable molecule that can diffuse inside the cells through water channels and acts as a signaling molecule (Tewari et al. 2021a, b). H_2O_2 regulates cellular functions related to growth, development, and their response to environmental cues (Mittler 2017). Studies suggest a direct relationship between the ROS-Ca^{2+} signaling pathway that enables cell-to-cell communication and the distant transmission of signals in plants (Steinhorst and Kudla 2013). Plant tissue may initiate an increase in $[Ca^{2+}]_{cyt}$ via sensing an influx from the extracellular space across the plasma membrane due to mechanical stimulation (Monshausen et al. 2009). This increased $[Ca^{2+}]_{cyt}$ activates RBOHC-dependent generation of ROS, following an amplification loop occurring between Ca^{2+} and RBOHC for regulating root hair development (Monshausen et al. 2007, 2009). Thus, Ca^{+2} and ROS work in coordination, and these might integrate via RBOHs and Ca^{+2} channels (TWO-PORE CHANNEL1), along with calcium-dependent protein kinases (CDPKs: CPK/CBL-CIPKs; and CBL-interacting protein kinases) (Choi et al. 2014; Gilroy et al. 2016). Elevation in transient Ca^{+2} has been regulated by Ca^{+2} sensors like calmodulin (CAM), calmodulin-like (CML), CDPK, and calcineurin B-like protein (CBL) (Asai et al. 2013; Bouche et al. 2005; McCormack et al. 2005). CDPKs have been shown to regulate the production of ROS

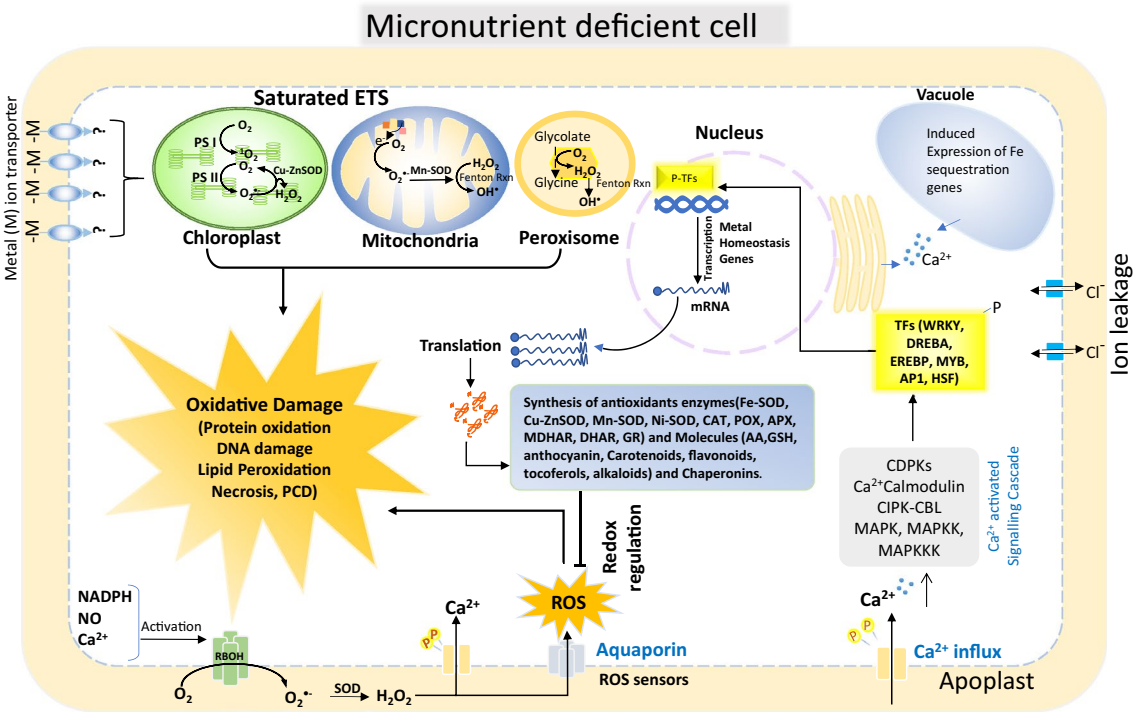


Fig. 6 ROS associated signaling events in micronutrient deficient plant cell. ROS are generated in three organelles: mitochondria, chloroplast, and peroxisomes. Under micronutrient deficiency, ROS are produced due to leakage of electron from oversaturated electron transport system (ETS), activation of respiratory burst oxidase homolog (RBOH) and weak antioxidant defense system. Elevated ROS may activate Ca^{2+} channel resulting in an influx of Ca^{2+} into cytosol from endoplasmic reticulum (ER), dictyosomes, and vacuole. Cytosolic Ca^{2+} eventually activate Ca^{2+}/calmodulin kinase, calcineurin B-like (CBL)-interacting protein kinases (CIPKs) to form CBL-CIPK signaling pathway, or mitogen activated protein kinase (MAPKs) pathway, which phosphorylates specific transcription factors [TFs: WRKY, EREBP (ethylene-responsive element-binding protein), DREBA (dehydration-responsive element-binding A), myb (myeloblastosis), AP1 (apetala1), HSF (heat shock factor)] for expressions of genes involved in biosynthesis of molecular antioxidants [ascorbic acid (AA), reduced glutathione (GSH), anthocyanins, carotenoids], antioxidant enzymes [SOD (superoxide dismutase, CAT (catalase), POD (peroxidase), APX (ascorbate peroxidase), MDHAR (monodehydroascorbate reductase), DHAR (dehydroascorbate reductase), GR (glutathione reductase), and HSPs (heat shock proteins). Excess ROS accumulation causes oxidative damage by lipid peroxidation (lipid Px), protein oxidation, DNA damage and induces programmed cell death (PCD) in plants

by phosphorylating the RBOHB protein. ROS-activated Ca^{2+} influx was observed to be upregulated by auxins in coordination with RBOH and PODs by the RSL4 (ROOT HAIR DEFECTIVE SIX-LIKE 4) transcription factor (Mangano et al. 2017).

The electron flow generated from water occurs through the chloroplastic ETS and via O_2^- to H_2O_2, and then reverting back to water will serve as a sink for excess electrons (water–water cycle) under stressed situations (Waszczak et al. 2018). Nitric oxide (NO), ROS, and sugars have been observed to function as activators of physiological responses to several nutritional deficiencies in plants (García et al. 2015). Under deficiency of micronutrients, plants may produce ROS due to: (1) the limited function electron transport system of the chloroplast and mitochondria, as several of the micronutrients, viz., Fe, Cu, and Mn, serve as integral components of the ETS (Grossman and Takahashi 2001; Tewari et al. 2007); (2) the weak antioxidant defense system (Fe, Cu, Zn, Mn, and Ni); and (3) the stimulation of ROS-producing enzymes RBOHs (Suzuki et al. 2011) as illustrated in Fig. 6.

ROS production may alter proteins functions through the oxidative-PTMs of different thiol groups, altering the cell redox status. Being an integral constituent of SODs, Fe, Mn, Cu, and Zn deficiency often accentuates the production of O_2^- (Tewari et al. 2013a, b; Tewari et al. 2006, 2019; Tewari et al. 2013a, b). Micronutrients, viz., Fe, Cu, and Mn, are integral cofactors of the proteins implicated in the ETS of chloroplasts and mitochondria (Broadley et al. 2012). Hence, the deficiency of any of these micronutrients suppresses the ETS consequently causing ROS accumulation. Nonetheless, Mo deficiency also enhanced production of O_2^- and H_2O_2 (Wu et al. 2017), along with changes (decrease or increase) in antioxidant enzymes such as SOD, POD, APX, and CAT (Li et al. 2001; Wu et al. 2017). Glutaredoxin, which may function as a redox sensor, and its transcript *GRX480* have also been proven to be upregulated under Mo deficiency, along with certain other transcripts such as heat shock factor (*HSF4*), and heat shock transcription factor (*HSFC1*) (Ide et al. 2011). BY-2 cells of tobacco accumulate H_2O_2 together with increased $[Ca^{2+}]_{cyt}$ concentration in deficiency of B (Koshiba et al. 2010, 2009). Thus, Ca^{2+} fluxes function in association with ROS (Steinhorst and Kudla 2013). The buildup of H_2O_2 causes the oxidation of the cysteine residues of the proteins. Oxidation of cysteine residue modifies the form and function of the proteins regulating signaling events (Mittler 2017; Mittler et al. 2022; Waszczak et al. 2018). Oxidation of cysteine residues present in the target proteins is catalyzed by peroxiredoxins (PRXS) and GSH peroxidases (GPXs) (Waszczak et al. 2018). These proteins have a high rate constant, and therefore they may act as ROS sensors (Waszczak et al. 2018). Upon oxidation, these sensing proteins (sensors) probably interact and help in oxidizing effector proteins, making a redox relay (Waszczak et al. 2018).

ROS perception by various receptors/sensors [CYSTEINE-RICH RECEPTOR-LIKE KINASES, namely CRKs (Bourdais et al. 2015) or HYDROGEN-PEROXIDE-INDUCED Ca^{2+} increases, HPCA1, an LRR kinase (Wu et al. 2020a, b), PRXS/GPXs (Waszczak et al. 2018), NUCLEOREXOXIN 1, NRX1 (Kneeshaw et al. 2017), GLUTAMATE RECEPTOR-LIKE, GLR (Fichman and Mittler 2021)] probably act in coordination with MAPKs cascade and various transcription factors (TFs) which control several cellular pathways (Li et al. 2022; Mittler 2002; Mittler et al. 2022). Sensing of H_2O_2 also relates to alterations in Ca^{2+} levels and calmodulin, leading to the activation of a Ca^{2+}-calmodulin kinase, which may also activate or suppress certain TFs (Mittler 2002). The control of gene expression by different TFs may induce defense pathways involved in ROS removal and heat shock proteins (HSPs) or suppress ROS-producing proteins (Mittler 2002). ROS generated due to mineral deficiency can stimulate the MAPK pathway (Jalmi and Sinha 2015), which may control TFs in a ROS-dependent way. Some TFs, like AP2/EREBP (RAP2.11, low K response), MYB (MYB114, anthocyanin biosynthesis; MYB75, anthocyanin metabolism), bHLH (TT8, anthocyanin biosynthesis), WRKY (WRKY50, WRKY51, represses jasmonate-mediated signaling; WRKY25, abiotic stress responses; WRKY7, plant defensive responses; WRKY63, regulation of ABA and drought stress responses), MADS (AGL32/TT16, proanthocyanidin biosynthesis) and NAC (ANAC092/ORE1, leaf senescence regulation) were found to be enhanced in glycolate oxidase (GO) over-expressing plants (Balazadeh et al. 2012). Modifications in transcriptional activities have been observed in GO plants soon after the induction of H_2O_2 formation in chloroplasts (Balazadeh et al. 2012). The FER-like iron deficiency-induced TF, FER/FIT, a positive transcriptional regulator of various Fe deficiency responses, has been found to be induced by Fe deficiency (Kobayashi and Nishizawa 2012). Expression of *FER/FIT*, is further enhanced by NO treatment (Graziano and Lamattina 2007; Meiser et al. 2011). Moreover, other *bHLH*TFs *[AtbHLH38, AtbHLH39,* and phosphorylated *Upstream Regulator of IRT1 (URI)]* act as necessary components of the signaling pathway of Fe deficiency in *A. thaliana* (Kim et al. 2019). The bZIP TF BdbZIP10, if overexpressed, protects the plants and callus tissue from oxidative damage, probably through the upregulation of antioxidant defense genes (Glover-Cutter et al. 2014). Recently, it has been proposed that H_2O_2 positively affects the expression of ZAT12, which in turn interacts with FIT. Subgroup IbbHLHs interact with FIT and form the IbbHLHs-FIT complex, which positively regulates FIT and different genes of Fe acquisition processes (FRO2 and IRT1). Fe taken up by FRO2 and IRT1 is utilized for

the catalytic function of CAT2 (von der Mark et al. 2021). However, under prolonged deficiency of Fe, high H_2O_2 accumulation attenuates expression of FRO2 and IRT1 (von der Mark et al. 2021). Moreover, Ca^{2+} affects IRT1-mediated import of Fe directly through the lipid-binding protein EHB1 triggering CBL-CIPK-mediated signaling and hence influencing the Fe-acquisition transcription factor FIT (Gratz et al. 2021).

Phosphorylation of proteins is a crucial mode of transducing signals in plants and is catalyzed by MAPKs (Li et al. 2022). MAPK cascades consist of three tier components such as MAPKKKs (for example: AtMEKK1, OMTK1, NDPK2), MAPKKs (for example, AtMKK2, AtMKK1, AtMKK3, AtMKK4/5, OsMKK4), and MAPKs (for example, AtMPK4, ZmSIMK1, AtMPK7, ZmMPK7, AtMPK3, ZmMPK3, AtMPK6, SIMK, SAMK) (Sinha et al. 2011). They catalyze the phosphorylation, starting from upstream receptors to downstream targets (Jalmi and Sinha 2015). Although their signaling processes seem complex, the MPK3, MPK4, and MPK6 of *A. thaliana* are key players in osmotic shock, oxidative stress, cold, and antipathogenic responses (Sinha et al. 2011)**.** These MAPKs are activated by both ligands and ROS (Mittler 2002) and thus, phosphorylation cascades are found to act both upstream and downstream of ROS formation (Asai et al. 2002). The MKK4-MPK3/6 module is known to produce ROS, working upstream of the RBOH function (Kovtun et al. 2000), and H_2O_2 adjusts the MPK3 and MPK6 (Kovtun et al. 2000). MPK6-2 responds to ROS signals during deficiency of Fe, and MPK6-2 regulates bHLH104 TF by phosphorylation (Kabir et al. 2021). Also, MPK6-2 overexpression in apple calli enhances their tolerance to deficiency of Fe by enhancing the expression of Fe-responsive genes (Kabir et al. 2021).

Conclusion and future prospective

Nutrient element deficiency produces ROS. ROS might act as a signaling molecule and activate the antioxidant defense system in plants by interacting with other key signaling partners such as RBOH, G-proteins, Ca^{2+}, MAPKs, and various TFs. Oxidative damage occurs when plants are unable to manage an excessive amount of ROS. The oxidative-PTMs of various enzymes related to antioxidant defense affect their functions. Ni and Cl are crucially important nutrient elements. However, plant responses, particularly ROS production and antioxidant defenses to Ni and Cl deficiencies, are unavailable. Although antioxidant responses under deficiency of Mn, Cu, Mo, and Zn are known, how perception and ROS signaling take place through targeting specific TFs by their activation or deactivation via Ca^{2+} or calmodulin kinase and/or MAPK cascade and various phosphatases is still unknown. Moreover, what different receptors and sensors work in the ROS signaling network under different micronutrient deficiencies is entirely unknown and needs thorough investigation.

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Data availability Data are available on request from the authors.

Declarations

Conflict of interest The authors declare that there has no conflict of interest.

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